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Inventor(s)	Zou; Zhenhua et al.

Handling reference time delivery failure in a time-sensitive network (TSN)

Abstract

Embodiments include methods for a user equipment (UE) to synchronize timing with a wireless network. Such methods include receiving, from a first node serving a first cell in the wireless network, a first message that includes an absolute reference time and a corresponding reference event within downlink, DL, transmissions by the first node; and determining whether the UE received, from the first node, a timing adjustment value from which the UE can estimate a propagation delay of the DL transmissions. Such methods include, based on determining that the UE did not receive the timing adjustment value, performing one of the following: transmitting, to the first node, an indication that the UE did not receive the timing adjustment value; or declaring radio link failure with respect to the first cell and establishing a connection with a second cell capable of providing the information included in the first message.

Inventors:	Zou; Zhenhua (Solna, SE), Shi; Nianshan (Järfälla, SE), Dudda; Torsten (Wassenberg, DE), Diachina; John Walter (Garner, NC)
Applicant:	Telefonaktiebolaget LM Ericsson (publ) (Stockholm, SE)
Family ID:	1000008821714
Assignee:	Telefonaktiebolaget LM Ericsson (publ) (Stockholm, SE)
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Primary Examiner: Soe; Kyaw Z

Attorney, Agent or Firm: Sage Patent Group

Background/Summary

TECHNICAL FIELD

(1) The present application relates generally to the field of wireless networks and more specifically to techniques for compensating for signal propagation delays between a wireless network and wireless devices, particularly when a wireless network and a wireless device are utilized to deliver highly accurate timing information from a time-sensitive network (TSN).

BACKGROUND

(2) Industry 4.0 is a term used to refer to a current trend of automation and data exchange in manufacturing. It can include concepts and/or technologies such as cyber-physical systems, the Internet of things (IoT), cloud computing, and cognitive computing. Industry 4.0 is also referred to as the fourth industrial revolution or 14.0" for short.

(3) One scenario or use case for Industry 4.0 is the so-called "smart factory." Within modular structured smart factories, cyber-physical systems monitor physical processes, create a virtual copy of the physical world, and make decentralized decisions. Over the Internet of Things (IoT), cyber-physical systems communicate and cooperate with each other, and with humans, in real-time both internally and across organizational services offered and used by participants of a value chain of which the smart factory is a part. Such smart factory environment environments are also referred to as Industrial Internet of Things (IIoT).

(4) There are four common principles associated with Industry 4.0. First, "interoperability" requires the ability to connect machines, devices, sensors, and people to communicate with each other via the IoT or, alternatively, the "Internet of People" (IoP). Second, "information transparency" requires information systems to have the ability to create a virtual copy of the physical world by enriching digital models (e.g., of a smart factory) actual with sensor data. For example, this can require the ability to aggregate raw sensor data to higher-value context information.

(5) Third, "technical assistance" requires assistance systems to be able to support humans by aggregating and visualizing information comprehensively for making informed decisions and solving urgent problems on short notice. This principle can also refer to the ability of cyber physical systems to physically support humans by conducting a range of tasks that are unpleasant, too exhausting, or unsafe for their human co-workers. Finally, cyber physical systems should have the ability to make decentralized decisions and to perform their tasks as autonomously as possible. In other words, only in the case of exceptions, interferences, or conflicting goals, should tasks be delegated to a higher level.

(6) These Industry 4.0 principles motivate various use cases that place many requirements on a network infrastructure. Use cases include simpler ones such as plant measurement to more difficult ones such as precise motion control in a robotized factory cell. To address these requirements, the IEEE 802.1 working group (particularly, task group TSN) has developed a Time Sensitive Networking (TSN) standard. TSN is based on the IEEE 802.3 Ethernet standard, a wired communication standard that is designed for "best effort" quality of service (QoS). TSN describes a collection of features intended to make legacy Ethernet performance more deterministic, including time synchronization, guaranteed low-latency transmissions, and improved reliability. The TSN

features available today can be grouped into the following categories (shown below with associated IEEE specifications): Time Synchronization (e.g., IEEE 802.1AS); Bounded Low Latency (e.g., IEEE 802.1Qav, IEEE 802.1Qbu, IEEE 802.1Qbv, IEEE 802.1Qch, IEEE 802.1Qcr); Ultra-Reliability (e.g., IEEE 802.1CB, IEEE 802.1Qca, IEEE 802.1Qci); Network Configuration and Management (e.g., IEEE 802.1Qat, IEEE 802.1Qcc, IEEE 802.1Qcp, IEEE 802.1CS).

(7) FIGS. 1-2 are block diagrams that respectively illustrate Centralized and Fully Centralized TSN configuration models, as specified in IEEE Std. 802.1Qbv-2015. Within a TSN network, the communication endpoints are called “Talker” and “Listener.” All the switches and/or bridges between a Talker and a Listener must support certain TSN features, such as IEEE 802.1AS time synchronization. A “TSN domain” includes all nodes that are synchronized in the network, and TSN communication is only possible within such a TSN domain.

(8) The communication between Talker and Listener is in streams. Each stream is based on data rate and latency requirements of an application implemented at both Talker and Listener. A Talker initializes a stream towards a Listener, and the TSN configuration and management features are used to set up the stream and to guarantee the stream's requirements across the network. Some TSN features require a central management entity called Centralized Network Configuration (CNC), as shown in FIG. 1. The CNC can use, for example, Netconf and YANG models to configure the switches in the network for each TSN stream. This also facilitates the use of time-gated queueing (defined in IEEE 802.1Qbv) that enables data transport in a TSN network with deterministic latency. With time-gated queueing on each switch, queues are opened or closed according to a precise schedule thereby allowing high-priority packets to pass through with minimum latency and jitter. Of course, packets must arrive at a switch ingress port before the gate is scheduled to be open.

(9) The fully centralized model shown in FIG. 2 also includes a Centralized User Configuration (CUC) entity used as a point of contact for Listener and Talker. The CUC collects stream requirements and endpoint capabilities from the devices and communicates with the CNC directly. Further details about TSN configuration are given in IEEE 802.1Qcc.

(10) Currently the fifth generation (“5G”) of cellular systems, also referred to as New Radio (NR), is being standardized within the Third-Generation Partnership Project (3GPP). NR is developed for maximum flexibility to support many different use cases. These include mobile broadband, machine type communication (MTC), ultra-reliable low-latency communications (URLLC), device-to-device (D2D), etc. The achievable latency and reliability performance of NR are important for these and other use cases related to IIoT and/or Industry 4.0. In order to extend NR applicability for such use cases, support for time synchronization in the 5G system via time sensitive network (TSN) has been defined in 3GPP TS 23.501 (v16.4.0).

(11) At a high level, the 5G network architecture consists of a Next Generation radio access network (NG-RAN) and a 5G core network (5GC). The NG-RAN includes various gNodeB's (gNBs, also referred to as base stations) serving cells by which wireless devices (also referred to as user equipment, or UEs) communicate. The gNBs can be connected to the 5GC via one or more NG interfaces and can be connected to each other via one or more Xn interfaces. Each of the gNBs can support frequency division duplexing (FDD), time division duplexing (TDD), or a combination thereof.

(12) FIG. 3 is a block diagram illustrating an exemplary division of the 5G network architecture into control plane (CP) and data (or user) plane (UP) functionality. For example, a UE (310) can communicate data packets to a device and/or application on an external network (e.g., the Internet) by sending them via the UE's serving gNB (321) in the NG-RAN (320) to a user plane function (UPF) in the 5GC. The UPF provides an interface between the 5GC and external networks.

(13) CP functionality can operate cooperatively with the UP functionality. CP functions shown in FIG. 3 include an access management function (AMF), a session management function (SMF), a network exposure function (NEF), a policy control function (PCF), a network repository function

(NRF), and a unified data management (UDM) function. The AMF can communicate with the RAN via an N2 logical interface, which can be carried over the NG interface from the gNB to the 5GC. Similarly, the UPF can communicate with the SMF via the N4 logical interface.

(14) To support IIoT uses cases, a 5G network (e.g., NG-RAN and 5GC) should be capable of delivering highly accurate timing information from an external TSN network to TSN endpoints connected to the 5G network, e.g., via UEs. FIG. 4 is a block diagram illustrating an exemplary arrangement for interworking between the 5G network architecture shown in FIG. 3 and an exemplary fully centralized TSN network architecture. Reference numbers used in FIG. 4 have the same meaning as in FIG. 3. In the following discussion, a device connected to the 5G network is referred to as 5G endpoint, and a device connected to the TSN domain is referred to as TSN endpoint. The arrangement shown in FIG. 4 includes a Talker TSN endpoint and a Listener 5G endpoint connected to a UE. In other arrangements, a UE can instead be connected to a TSN network comprising at least one TSN bridge and at least one TSN endpoint. In this configuration, the UE can be part of a TSN-5G gateway.

(15) The TSN can include a grandmaster clock (TSN GM) that serves as the definitive timing source for TSN endpoints. At a high level, the 5G network in FIG. 4 should appear to the connected TSN as a switch or bridge that delivers the TSN GM timing to the connected endpoints in compliance with the requirements in IEEE 802.1AS. However, the 5G network does not use the TSN GM as its own timing source, but instead relies on a 5G system clock (5GSC) that is distributed among the various network nodes or functions. As such, one or more timing relationships between TSN GM and 5GSC may need to be determined and/or derived to facilitate transit of the TSN GSM to the connected end station in a compliant manner.

(16) At a high level, the time synchronization solution defined in 3GPP TS 23.501 only requires NG-RAN nodes (e.g., gNBs) to be synchronized to the 5G network reference time (i.e., based on 5GSC) while TSN GM timing is delivered to UEs and endpoints transparently through the 5G network using higher-layer generalized precision time protocol (gPTP) signaling. For 5GSC synchronization, a UE relies on its serving gNB providing reference time periodically, either via broadcast or unicast signaling. The nominal periodicity $T_{sub.n}$ of gNB reference time delivery is left to network implementation. However, $T_{sub.n}$ can reflect the UE clock stability and gNB clock stability in relation to the 5G GM clock used as the basis of the 5G reference time, etc.

(17) Due to various reasons, the UE's connection to the serving gNB can suffer temporary outages, causing a radio link failure (RLF) in the UE. It is currently unclear how the UE can maintain sufficiently accurate synchronization to 5G network reference time during those outages.

SUMMARY

(18) Embodiments of the present disclosure provide specific improvements to time-sensitive networking (TSN) in a wireless environment, such as by facilitating solutions to overcome the exemplary problems summarized above and described in more detail below

(19) Embodiments include methods (e.g., procedures) for a user equipment (UE, e.g., wireless device, IoT device) to synchronize timing with a wireless network (e.g., E-UTRAN, NG-RAN).

(20) These exemplary methods can include receiving, from a first node serving a first cell in the wireless network, a first message that includes an absolute reference time and a corresponding reference event within downlink (DL) transmissions by the first node. These exemplary methods can also include determining whether it received, from the first node, a timing adjustment value from which the UE can estimate a propagation delay of the DL transmissions from the first node to the UE. These exemplary methods can also include, based on determining that the UE did not receive the timing adjustment value, performing one of various operations. In some embodiments, the UE can transmit, to the first node, an indication that the UE did not receive the timing adjustment value. In other embodiments, the UE can declare radio link failure (RLF) with respect to the first cell and establish a connection with a second cell that is capable of providing the information included in the first message.

- (21) In some embodiments, these exemplary methods can also include receiving, from the first node, a retransmission of the timing adjustment value after transmitting the indication.
- (22) In some embodiments, the timing adjustment value is a timing advance (TA) command that indicates an amount of timing adjustment that the UE should apply to UL transmissions. In some embodiments, the determining operations can include determining whether the UE received a medium access control (MAC) control element (CE) comprising the timing adjustment value during a most recent first duration and/or a second duration after receiving the first message.
- (23) In some embodiments, these exemplary method can also include receiving, from the first node, a timer value associated with a timing advance group (TAG) that includes the first cell; and initiating a timer based on the timer value upon receiving a previous timing adjustment value from the first node. In such embodiments, determining that the UE did not receive the timing adjustment value can be based on expiration of the timer. Also, in some embodiments, transmitting the indication that the UE did not receive the timing adjustment value can be responsive to expiration of the timer. Furthermore, in some embodiments, declaring RLF can be based on the timer being initiated and expiring a plurality of consecutive times without receiving the timing adjustment value.
- (24) In some embodiments, these exemplary methods can also include, based on determining that the UE did receive the timing adjustment value, selectively using the timing adjustment value for estimating the propagation delay of the DL transmissions. In some embodiments, selectively using can include using the timing adjustment value only when the value of a flag, received with the TA command in the MAC CE, is different from a most recent value of the flag received with a most recent TA command.
- (25) In some embodiments, these exemplary methods can also include sending, to the first node, an indication of a UE preference for receiving reference time information from the wireless network. In such embodiments, receiving the first message can be in response to the indication.
- (26) In some embodiments, establishing a connection with a second cell that is capable of providing the information included in the first message can include receiving information broadcast by the second cell, and determining that the broadcast information includes a field reserved for delivery of reference time information.
- (27) In other embodiments, establishing a connection with a second cell that is capable of providing the information included in the first message can include sending, to a second node serving the second cell, a reestablishment request message comprising a request for reference time information; and receiving, from the second node, a unicast reestablishment message comprising the requested reference time information.
- (28) In some embodiments, the first node, the first cell, and the second cell can be part of a standalone non-public network (SNPN). In other embodiments, the first node can be part of a public network integrated non-public network (PNI-NPN). In such embodiments, the UE can be part of a closed access group (CAG) associated with the PNI-NPN, and the first cell and the second cell are associated with the CAG.
- (29) Other embodiments include methods (e.g., procedures) for a first node (e.g., base station, eNB, gNB, etc., or component thereof) serving a first cell in a wireless network (e.g., E-UTRAN, NG-RAN) to synchronize timing with a UE.
- (30) These exemplary methods can include transmitting, to the UE, a first message that includes an absolute reference time and a corresponding reference event within DL transmissions by the first node. These exemplary methods can also include periodically transmitting, to the UE, respective timing adjustment values from which the UE can estimate a propagation delay of the DL transmissions from the first node to the UE. These exemplary methods can also include, based on determining that the UE did not receive one of the periodic transmissions of the respective timing adjustment values, retransmitting the non-received timing adjustment value to the UE.
- (31) In some embodiments, the timing adjustment value can be a TA command that indicates an

amount of timing adjustment that the UE should apply to UL transmissions. In addition, the timing adjustment value can be transmitted in a MAC CE.

(32) In some embodiments, these exemplary methods can also include sending, to the UE, a timer value associated with a TAG that includes the first cell. The timer value can be related to a period between successive transmissions of respective timing adjustment values. In some embodiments, these exemplary methods can also include receiving, from the UE, an indication that the UE did not receive the timing adjustment value during a duration corresponding to the timer value. The determination can be based on receiving this indication.

(33) In some embodiments, each timing adjustment value can be transmitted in a MAC CE together with a flag. In some embodiments, a first value of the flag can indicate that the timing adjustment value is different from a most recently transmitted timing adjustment value, and a second value of the flag can indicate that the timing adjustment value is the same as the most recently transmitted timing adjustment value.

(34) In some embodiments, these exemplary methods can also include receiving, from the UE, an indication of the UE's preference for receiving reference time information from the wireless network. In such embodiments, the first message can be sent in response to this indication.

(35) In some embodiments, the first node and the first cell are part of an SNPN. In other embodiments, the first node can be part of a PNI-NPN, the UE can be part of a CAG associated with the PNI-NPN, and the first cell is associated with the CAG. In such embodiments, these exemplary methods can also include transmitting the first message in one or more further cells associated with the CAG.

(36) Other embodiments include UEs (e.g., wireless devices, IoT devices, etc. or components thereof) and network nodes (e.g., base stations, eNBs, gNBs, etc., or components thereof) configured to perform operations corresponding to any of the exemplary methods described herein. Other embodiments include non-transitory, computer-readable media storing program instructions that, when executed by processing circuitry, configure such UEs and network nodes to perform operations corresponding to any of the exemplary methods described herein. These and other embodiments described herein can enable a UE to maintain accurate reference time synchronization even when radio link conditions towards its serving RAN node degrade to failure. This can benefit time-sensitive applications that rely on the UE remaining tightly synchronized with 5GSC.

(37) These and other objects, features, and advantages of embodiments of the present disclosure will become apparent upon reading the following Detailed Description in view of the Drawings briefly summarized below.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a block diagram illustrating a Centralized Time-Sensitive Networking (TSN) configuration model, as specified in IEEE 802.1Qbv-2015.
- (2) FIG. 2 is a block diagram illustrating a Fully Centralized TSN configuration model, as specified in IEEE 802.1Qbv-2015.
- (3) FIG. 3 is a block diagram illustrating an exemplary control plane (CP) and a data (or user) plane (UP) architecture of an exemplary 5G wireless network.
- (4) FIG. 4 is a block diagram illustrating an exemplary arrangement for interworking between the 5G network architecture shown in FIG. 3 and an exemplary fully centralized TSN network architecture.
- (5) FIG. 5 is a high-level block diagram of an exemplary arrangement in which a 5G network delivers timing references from a time-sensitive network (TSN) to TSN end stations connected to

the 5G network, according to various exemplary embodiments of the present disclosure.

(6) FIG. 6 illustrates an exemplary reference time update procedure between a UE and a serving network node (e.g., gNB).

(7) FIG. 7 shows an ASN.1 data structure for an exemplary ReferenceTimeInfo information element (IE) used to provide the reference time to a UE.

(8) FIG. 8 shows an ASN.1 data structure for a UEAssistanceInformation message by which a UE can indicate preference(s) for receiving 5G reference time information.

(9) FIG. 9 shows an exemplary timing advance (TA) medium access control (MAC) control element (CE) in which a reserved (“R”) bit has been purposed into a flag (“F”), according to various embodiments of the present disclosure.

(10) FIG. 10 is a flow diagram of an exemplary method (e.g., procedure) for a UE (e.g., wireless device), according to various embodiments of the present disclosure.

(11) FIG. 11 shows a flow diagram of an exemplary method (e.g., procedure) for a network node (e.g., base station, eNB, gNB, etc.), according to various embodiments of the present disclosure.

(12) FIG. 12 illustrates an exemplary wireless network, according to various embodiments of the present disclosure.

(13) FIG. 13 illustrates an exemplary UE, according to various embodiments of the present disclosure.

(14) FIG. 14 is a block diagram illustrating an exemplary virtualization environment usable for implementing various embodiments of the present disclosure.

(15) FIGS. 15-16 are block diagrams of exemplary communication systems and/or networks, according to various embodiments of the present disclosure.

(16) FIGS. 17-20 are flow diagrams illustrating exemplary methods (e.g., procedures) implemented in a communication system, according to various embodiments of the present disclosure.

DETAILED DESCRIPTION

(17) Some of the embodiments contemplated herein will now be described more fully with reference to the accompanying drawings. Other embodiments, however, are contained within the scope of the subject matter disclosed herein, the disclosed subject matter should not be construed as limited to only the embodiments set forth herein; rather, these embodiments are provided by way of example to convey the scope of the subject matter to those skilled in the art.

(18) Generally, all terms used herein are to be interpreted according to their ordinary meaning in the relevant technical field, unless a different meaning is clearly given and/or is implied from the context in which it is used. All references to a/an/the element, apparatus, component, means, step, etc. are to be interpreted openly as referring to at least one instance of the element, apparatus, component, means, step, etc., unless explicitly stated otherwise. The steps of any methods and/or procedures disclosed herein do not have to be performed in the exact order disclosed, unless a step is explicitly described as following or preceding another step and/or where it is implicit that a step must follow or precede another step. Any feature of any of the embodiments disclosed herein can be applied to any other embodiment, wherever appropriate. Likewise, any advantage of any of the embodiments can apply to any other embodiments, and vice versa. Other objects, features, and advantages of the enclosed embodiments will be apparent from the following description.

(19) Furthermore, the following terms are used throughout the description given below: Radio Node: As used herein, a “radio node” can be either a “radio access node” or a “wireless device.” Radio Access Node: As used herein, a “radio access node” (or equivalently “radio network node,” “radio access network node,” or “RAN node”) can be any node in a radio access network (RAN) of a cellular communications network that operates to wirelessly transmit and/or receive signals. Some examples of a radio access node include, but are not limited to, a base station (e.g., a New Radio (NR) base station (gNB) in a 3GPP Fifth Generation (5G) NR network or an enhanced or evolved Node B (eNB) in a 3GPP LTE network), base station distributed components (e.g., CU and DU), a high-power or macro base station, a low-power base station (e.g., micro, pico, femto, or

home base station, or the like), an integrated access backhaul (IAB) node, a transmission point, a remote radio unit (RRU or RRH), and a relay node. Core Network Node: As used herein, a “core network node” is any type of node in a core network. Some examples of a core network node include, e.g., a Mobility Management Entity (MME), a serving gateway (SGW), a PDN Gateway (P-GW), a Policy and Charging Rules Function (PCRF), an access and mobility management function (AMF), a session management function (SMF), a user plane function (UPF), a Charging Function (CHF), a Policy Control Function (PCF), an Authentication Server Function (AUSF), or the like. Wireless Device: As used herein, a “wireless device” (or “WD” for short) is any type of device that has access to (i.e., is served by) a cellular communications network by communicate wirelessly with network nodes and/or other wireless devices. Communicating wirelessly can involve transmitting and/or receiving wireless signals using electromagnetic waves, radio waves, infrared waves, and/or other types of signals suitable for conveying information through air. Unless otherwise noted, the term “wireless device” is used interchangeably herein with “user equipment” (or “UE” for short). Some examples of a wireless device include, but are not limited to, smart phones, mobile phones, cell phones, voice over IP (VoIP) phones, wireless local loop phones, desktop computers, personal digital assistants (PDAs), wireless cameras, gaming consoles or devices, music storage devices, playback appliances, wearable devices, wireless endpoints, mobile to stations, tablets, laptops, laptop-embedded equipment (LEE), laptop-mounted equipment (LME), smart devices, wireless customer-premise equipment (CPE), mobile-type communication (MTC) devices, Internet-of-Things (IoT) devices, vehicle-mounted wireless terminal devices, etc. Network Node: As used herein, a “network node” is any node that is either part of the radio access network (e.g., a radio access node or equivalent name discussed above) or of the core network (e.g., a core network node discussed above) of a cellular communications network. Functionally, a network node is equipment capable, configured, arranged, and/or operable to communicate directly or indirectly with a wireless device and/or with other network nodes or equipment in the cellular communications network, to enable and/or provide wireless access to the wireless device, and/or to perform other functions (e.g., administration) in the cellular communications network.

(20) Note that the description given herein focuses on a 3GPP cellular communications system and, as such, 3GPP terminology or terminology similar to 3GPP terminology is oftentimes used. However, the concepts disclosed herein are not limited to a 3GPP system. Furthermore, although the term “cell” is used herein, it should be understood that (particularly with respect to 5G NR) beams may be used instead of cells and, as such, concepts described herein apply equally to both cells and beams.

(21) As briefly mentioned above, to maintain synchronization with a 5G system clock (5GSC), a UE relies on its serving gNB to provide reference time periodically, either via broadcast or unicast signaling. Due to various reasons, however, the UE's connection to the serving gNB might be lost temporarily. It is currently unclear how the UE can maintain sufficiently accurate synchronization to 5G network reference time during those outages This is discussed in more detail below after the following description of TSN and various aspects of 5G networks.

(22) FIG. 5 is a high-level block diagram of an exemplary arrangement in which a 5G network delivers timing references from a TSN network to TSN end stations connected to the 5G network. The TSN source network is shown as a TSN Working Domain that includes a TSN GM, an end station, and a TSN Switch. In this exemplary arrangement, the 5G network is integrated with the external TSN network as a TSN bridge, such as shown in other figures described above.

Furthermore, in this arrangement, the 5G network can be modelled as an IEEE 802.1AS compliant entity; for TSN synchronization, the end-to-end 5G network can be considered as an IEEE 802.1AS “time-aware system”.

(23) However, only the TSN Translators (TTs) at the edges of the 5G network need to support the IEEE 802.1AS operations. This includes a network side TSN translator (NW-TT) at the user plane function (UPF) and a device-side TSN translator (DS-TT) at the UE. Ingress timestamping (“TSi”)

is performed by the NW-TT through an external TSN clock (e.g., timing) signal is received by from the TSN Working Domain. Likewise, egress timestamping (“TSe”) is performed by the DS-TT when that TSN clock signal arrives at a UE. In addition to ingress and egress timestamping, the TTs can support other 802.1AS functions such as generalized precision time protocol (gPTP), Best Master Clock Algorithm (BMCA), rateRatio, etc.

(24) More specifically, upon reception of a downlink gPTP message, the NW-TT makes an ingress timestamp (TSi) for each gPTP event (Sync) message. The UPF then forwards the gPTP message from TSN network to the UEs via all UPF-terminated PDU sessions that the UEs have established to the TSN network. All gPTP messages are transmitted on a quality of service (QoS) flow that complies with the residence time upper bound requirement specified in IEEE 802.1AS. The UE receives the gPTP messages and forwards them to the DS-TT. The DS-TT then makes an egress timestamp (TSe) for the gPTP event (Sync) messages for the TSN domain (e.g., the endpoints).

(25) The difference between TSi and TSe reflects the residence time of the gPTP message within the 5G network, expressed in 5GSC time. Put differently, if gPTP message indicating TSN time “X” is stamped with 5G system clock (5GSC) time “Y” at ingress and 5GSC time “Z” at egress, the end stations can adjust TSN time “X” delivered to the end station by the residence time Z-Y. More specifically, the DS-TT calculates and adds the measured residence time between the TTs into the Correction Field (CF) of each gPTP event (Sync) message. As such, the relative accuracy of the 5G residence time measured between ingress and egress is essential for accurate TSN GM clock timing information delivered over 5G networks.

(26) In the 5G network, the UE, the gNB, the UPF, the NW-TT, and the DS-TT are synchronized with a grandmaster 5GSC (“5G GM” in FIG. 5), either directly or indirectly. In general, the 5GSC is made available to all UP nodes in the 5G network via a PTP-compatible transport network. Likewise, 5GSC is made available to UEs via signaling of absolute timing of radio frames. In FIG. 5, solid lines are used to denote flow of 5GSC synchronization between respective synchronization master (“M”) and slave (“S”) elements in the 5G network. Likewise, dashed lines are used to denote flow of TSN GM synchronization between respective synchronization master (“M”) and slave (“S”) elements in the TSN domain. In general, the two synchronization processes can be independent from each other and the gNB only needs to be synchronized to the 5GSC.

(27) To summarize, the time synchronization solution defined in 3GPP TS 23.501 (v16.4.0) only requires NG-RAN nodes (e.g., gNBs) to be synchronized to the 5G network reference time (i.e., based on 5GSC) while TSN GM timing is delivered to UEs and endpoints transparently through the 5G network using gPTP signaling. For 5GSC synchronization, a UE relies on its serving gNB providing reference time periodically, either via broadcast or unicast signaling. For example, reference time information can be delivered via broadcast in SI block SIB9 or via unicast via the RRC message DownlinkInformationTransfer. To support time sensitive communications, a reference time granularity of 10 ns is defined in 3GPP TS 23.501.

(28) FIG. 6 illustrates an exemplary reference time update procedure between a UE and a serving gNB. In SFNx, the gNB transmits a reference time message (e.g., SIB9 or unicast) with a 5G reference time value (t.sub.R) and a corresponding reference event on the radio interface for the cell. In the example shown in FIG. 6, the reference event is the end of SFNz that is subsequent to SFNx. For SIB9, the value of SFN.sub.z is implicitly indicated as the SFN boundary at or immediately after the ending boundary of the SI-window in which SIB9 is transmitted. When the reference time is sent in a unicast message, the reference cell of the time at the ending boundary of the SFN indicated by the referenceSFN field is the UE's PCell. The UE receives the reference time message in advance of the reference event and subsequently synchronizes its internal clock with the reference time t.sub.R that occurs at the end of SFNz.

(29) NG-RAN nodes, such as gNBs, can include a central (or centralized) unit (CU or gNB-CU) and one or more distributed (or decentralized) units (DU or gNB-DU). CUs are logical nodes that host higher-layer protocols and perform various gNB functions such controlling the operation of

DUs. Each DU is a logical node that hosts lower-layer protocols and can include, depending on the functional split, various subsets of the gNB functions. In the context of providing reference time to UEs, the SIB and RRC unicast messages are generated by a CU and, preferably, the reference event is referenced on the radio interface provided by the DU. Thus, a DU can overwrite SIB9 for broadcast delivery. For unicast delivery, a DU reports 5GSC time at reference event (e.g., SFN) to CU, which generates the unicast RRC message that includes the reported timing relationship.

(30) FIG. 7 shows an ASN.1 data structure for an exemplary ReferenceTimeInfo information element (IE) used to provide the reference time to a UE. The IE includes a time field that is defined as a ReferenceTime structure that includes fields for days, seconds, milliseconds (ms), and tens of nanoseconds (ns). The IE also includes a referenceSFN that defines the reference event on the radio interface. In addition, the uncertainty field reflects a level of uncertainty in the provided timing relationship. This includes two factors: 1) the accuracy with which a gNB implementation can ensure that the indicated reference time corresponding to reference point will reflect the actual time when that reference point occurs at the antenna reference point, and 2) the accuracy with which the 5G reference time is acquired by the gNB. The actual uncertainty is 25 ns multiplied by the integer value in this field. If this field is absent, the uncertainty is unspecified.

(31) If the timeInfoType is not included in the IE, the time field indicates the GPS time and the origin of the time field is 00:00:00 on Gregorian calendar date 6 Jan. 1980 (i.e., start of GPS time). If timeInfoType is set to localClock, the origin of the time is unspecified.

(32) In addition, to support time-sensitive communications in Rel-16, the UE Assistance Information procedure has been extended to allow a UE to indicate its preference(s) for receiving 5G reference time information. FIG. 8 shows an ASN.1 data structure for a UEAssistanceInformation message by which a UE can indicate these preferences. In particular, the reference TimeInfoPreference IE is a Boolean value that indicates when the UE's preference for reference time information has changed, i.e., from true to false or from false to true. The UE cannot send another indication with the same value. Once UE send the preference request, UE relies on periodic gNB broadcast/unicast to refresh its reference time and should no longer re-send the request to the network.

(33) One problem that arises in determining these timing relationships is the radio frequency (RF) propagation delay (PD) of the signal from the gNB to the UE, which is proportional to the distance between gNB and UE. For example, a UE that is 300m distant from the gNB antenna will experience a propagation delay of approximately one microsecond. As such, even if the gNB provides the UE with a 5GSC time, the TSN time derived by the UE may be inaccurate (e.g., offset) by the amount of the PD. Put differently, a TSN-5GSC timing relationship is only accurate up to the point of transmission of the TSN message by the gNB, e.g., at the gNB's transmission antenna(s). Furthermore, UEs may experience PDs that vary with time as the distance to the serving gNB changes and/or the serving gNB changes. Likewise, different UEs served by the same gNB may experience different PDs.

(34) Since the most stringent end-to-end synchronization requirement for time-sensitive applications is also one microsecond, the propagation delay from the gNB to the UE must be considered. The 3GPP Timing Advance command (see 3GPP TS 38.133) is used for UE uplink (UL) transmission synchronization. This may be needed due to changes in the UE propagation environment and/or distance between the UE and the serving base station (e.g., gNB). At connection setup, an absolute timing correction is communicated to a UE using a medium access control (MAC) random access response (RAR) element. After connection setup, a relative timing correction can be sent to a UE using a MAC control element (CE).

(35) The downlink (DL) propagation delay (PD) can be estimated for a given UE by (a) summing the TA value indicated by the initial absolute TA value and all subsequent relative TA values, and (b) taking some portion of the resulting total TA value to represent DL-only delays. For example, 50% could be used assuming the DL and UL propagation delays are essentially the same. The

estimated PD can then be used to understand time synchronization dynamics, e.g., for accurately tracking and/or compensating the value of a 5GSC at the UE side relative to the value of that clock in some other network node.

(36) The radio resource control (RRC) layer of the control plane (CP) controls UE mobility between cells in the NG-RAN as well as between NG-RAN and previous generation Evolved UTRAN (E-UTRAN). After a UE is powered on it will be in the RRC_IDLE state until an RRC connection is established with the network, at which time the UE will transition to RRC_CONNECTED state (e.g., where data transfer can occur). The UE returns to RRC_IDLE after the connection with the network is released. In RRC_IDLE state, the UE does not belong to any cell, no RRC context has been established for the UE (e.g., in NG-RAN), and the UE is out of UL synchronization with the network. Even so, a UE in RRC_IDLE state is known in the core network (e.g., 5GC) and has an assigned IP address.

(37) Furthermore, in RRC_IDLE state, the UE's radio is active on a discontinuous reception (DRX) schedule configured by upper layers. During DRX active periods (also referred to as “On durations”), an RRC_IDLE UE receives system information (SI) broadcast by a serving cell, performs measurements of neighbor cells to support cell reselection, and monitors a paging channel for pages from the core network via a RAN node (e.g., gNB) serving the cell in which the UE is camping. A UE must perform a random-access procedure to move from RRC_IDLE to RRC_CONNECTED state. In RRC_CONNECTED, the cell serving the UE is known and an RRC context is established for the UE in the serving RAN node, such that the UE and the RAN node can communicate.

(38) A common mobility procedure for UEs in RRC_CONNECTED state (e.g., with an active connection) is handover (HO) between cells. A UE is handed over from a source or serving cell, provided by a source node, to a target cell provided by a target node. Seamless handovers are a key feature of 3GPP technologies. Successful handovers ensure that the UE moves around in the coverage area of different cells without causing too many interruptions in data transmission.

(39) The NR RRC layer also includes an RRC_INACTIVE state in addition to RRC_IDLE and RRC_CONNECTED. The UE's RRC connection can be suspended to RRC_INACTIVE and then later resumed (i.e., to RRC_CONNECTED state). The suspended UE's RRC context is stored by the NG-RAN, which can page the UE while in RRC_INACTIVE.

(40) A dual connectivity (DC) framework was introduced in Long Term Evolution (LTE) Rel-12. Dual connectivity refers to a mode of operation in which a UE, in RRC_CONNECTED state, consumes radio resources provided by at least two different network points connected to one another with a non-ideal backhaul. In LTE, these two network points may be referred to as a “Master eNB” (MeNB) and a “Secondary eNB” (SeNB). More generally, master node (MN), anchor node, and MeNB can be used interchangeably, and the terms secondary node (SN), booster node, and SeNB can be used interchangeably. DC can be viewed as a special case of carrier aggregation (CA) in which the aggregated carriers (or cells) are provided by network nodes that are physically separated and not connected via a robust, high-capacity connection.

(41) In LTE DC, a UE is configured with a Master Cell Group (MCG) associated with the MeNB and a Secondary Cell Group (SCG) associated with the SeNB. Each of the CGs is a group of serving cells that includes one MAC entity, a set of logical channels with associated RLC entities, a primary cell (PCell), and optionally one or more secondary cells (SCells). The term “Special Cell” (or “SpCell” for short) refers to the PCell of the MCG or the PCell of the SCG depending on whether the UE's MAC entity is associated with the MCG or the SCG, respectively. In non-DC operation (e.g., CA), SpCell refers to the PCell. An SpCell is always activated and supports physical UL control channel (PUCCH) transmission and contention-based random access by UEs.

(42) The MeNB provides system information (SI) and terminates the control plane connection towards the UE and, as such, is the controlling node of the UE, including handovers to and from SeNBs. For example, the MeNB terminates the connection between the eNB and the Mobility

Management Entity (MME) for the UE. An SeNB provides additional radio resources (e.g., bearers) for radio resource bearers include MCG bearers, SCG bearers, and split bearers that have resources from both MCG and SCG. Data radio bearers (DRBs) carry user plane data while signaling radio bearers (SRBs, e.g., SRB0-3) carry control plane information (e.g., RRC and NAS messages).

(43) The reconfiguration, addition and removal of SCells can be performed by RRC. When adding a new SCell, dedicated RRC signaling is used to send the UE all required SI of the SCell, such that UEs need not acquire SI directly from the SCell broadcast. It is also possible to support CA in either or both of MCG and SCG. In other words, either or both of the MCG and the SCG can include multiple cells working in CA.

(44) DC is also envisioned as an important feature for 5G networks. Several DC (or more generally, multi-connectivity) scenarios have been considered for NR. These include NR-DC that is similar to LTE-DC discussed above, except that both the MN and SN (e.g., gNBs) employ the NR radio interface to communicate with the UE. In addition, various multi-RAT DC (MR-DC) scenarios have been considered, whereby a compatible UE can be configured to utilize resources provided by two different nodes, one providing E-UTRA/LTE access and the other one providing NR access. One node acts as the MN (e.g., providing MCG) and the other as the SN (e.g., providing SCG), with the MN and SN being connected via a network interface and at least the MN being connected to a core network (e.g., EPC or 5GC).

(45) In MR-DC, a SN is not required to broadcast system information (SI) other than for radio frame timing and system frame number (SFN). The UE's MN provides SI for initial configuration by dedicated RRC signaling to the UE. The UE acquires SN radio frame timing and SFN of the SCG from the PSCell, i.e., PSS/SSS and MIB if the SN is an eNB or PSS/SSS and PBCH if the SN is a gNB.

(46) All MR-DC arrangements support split SRB1 and split SRB2 but do not support split SRB0 and split SRB3. Split SRBs can be configured by the MN via an SN Addition or SN Modification procedures, with SN configuration part provided by the SN. A UE can be configured with both SRB3 and a split SRB simultaneously, e.g., SRB3 and the SCG leg of split SRB can be independently configured. For the split SRB, the selection of DL transmission path depends on network implementation. For UL, the UE is configured via MN RRC signaling whether to use MCG path or duplicate the transmission on both MCG and SCG.

(47) When CA is deployed, frame timing and SFN are aligned across cells that can be aggregated, or a timing offset between the PCell/PSCell and an SCell (e.g., number of slot) is configured for the UE. In CA, a UE may simultaneously receive or transmit on one or multiple CCs depending on its TA capabilities. A UE with only single TA capability for CA can simultaneously receive and/or transmit on multiple CCs corresponding to multiple serving cells that share the same TA (i.e., grouped in one TAG). A UE with multiple TA capability for CA can simultaneously receive and/or transmit on multiple CCs corresponding to multiple serving cells with different TAs (i.e., multiple serving cells grouped in multiple TAGs). The NG-RAN ensures that each TAG contains at least one serving cell. Finally, a non-CA capable UE can receive on a single CC and transmit on a single CC corresponding to one serving cell only, i.e., one serving cell in one TAG.

(48) 3GPP specifications also define Non-Public Networks (NPN) that are for non-public use. These can be deployed as a Stand-alone Non-Public Network (SNPN) when not relying on network functions provided by a public land mobile network (PLMN). An SNPN is identified by a PLMN ID and (network ID) NID broadcast in SIB1. An SNPN-capable UE supports the SNPN access mode. When the UE is set to operate in SNPN access mode, the UE only selects and registers with SNPNs. When the UE is not set to operate in SNPN access mode, the UE performs normal PLMN selection procedures. UEs operating in SNPN access mode only (re)select cells within the selected/registered SNPN and a cell can only be considered as suitable if the PLMN and NID broadcast by the cell matches the selected/registered SNPN.

(49) Alternately, NPNs can be deployed as a Public Network Integrated (PNI) NPN when relying on functions provided by a PLMN. For PNI-NPNs, Closed Access Groups (CAGs) identify groups of subscribers who are permitted to access one or more cells associated with the CAG. A CAG is identified by a CAG identifier broadcast in SIB 1. A CAG-capable UE can be configured with the following per PLMN: Allowed CAG list containing the CAG identifiers which the UE is allowed to access; and CAG-only indication if the UE is only allowed to access 5GS via CAG cells.

The UE checks the suitability of CAG cells based on the Allowed CAG list provided by upper layers. When the UE is configured with a CAG-only indication, only CAG Member Cells can be suitable. A non-suitable cell can be acceptable though if the UE is configured with a CAG-only indication for one of the PLMN broadcast by the cell.

(50) Roaming and access restriction information for a UE includes information on restrictions to be applied for subsequent mobility action during operation in CM-CONNECTED state. This information may be provided and/or updated by the AMF in 5GC, and can include a forbidden radio access technology (RAT), a forbidden area, and service area restrictions as specified in 3GPP TS 23.501. This information can also include a serving PLMN/SNPN, a list of equivalent PLMNs, and/or PNI-NPN mobility restrictions (i.e., list of Allowed CAGs and whether the UE can also access PLMN cells).

(51) As discussed above, for 5GSC synchronization, a UE relies on its serving gNB to deliver (or “refresh”) reference time periodically, either via broadcast or unicast signaling. The nominal periodicity $T_{sub.n}$ of gNB reference time delivery is left to network implementation. However, $T_{sub.n}$ can reflect the UE clock stability and gNB clock stability in relation to the 5G GM clock used as the basis of the 5G reference time, etc. Due to various reasons, the UE's connection to the serving gNB might be lost temporarily. For example, temporary blockage of radio signals might lead to a radio link failure (RLF) in the UE. It is currently unclear how the UE can maintain sufficiently accurate synchronization to 5G network reference time during those outages.

(52) Embodiments of the present disclosure address these and other needs, problems, and/or difficulties by techniques for addressing a UE's failure to receive timely refresh of reference time information from a network node (e.g., gNB) in a wireless network (e.g., NG-RAN). A successful refresh includes a UE receiving both a first value of uncompensated 5G reference time (i.e., value of the 5G reference time maintained by the UE's serving gNB) and a second value (e.g., TA) from which the UE can estimate the DL propagation delay for compensating the received reference time value. The UE can determine that it failed to receive either the first value or the second value during a first duration, or that it received the first value and the second value more than a second duration apart. Based on either of these determinations, the UE can inform the serving gNB about the condition, trigger radio link failure (RLF), and re-establish connectivity towards another gNB that supports synchronization with 5G reference time. In this manner, the UE can maintain accurate reference time synchronization even when radio link conditions towards its serving gNB degrade to failure, which facilitates time sensitive networking (TSN) availability for higher-layer applications.

(53) As discussed above, accurate UE synchronization with network reference time requires the UE to receive a TA MAC control element (CE), from which UE can deduce how much it needs to adjust its UL transmission timing and determine a corresponding DL propagation delay based on an assumption of UL/DL symmetry. After transmitting a TA MAC CE to a UE, the network assumes that the UE adjusts its UL transmission timing accordingly.

(54) In the description of various embodiments, the term “TA MAC CE” generally refers to a MAC CE transmitted by the network (e.g., serving gNB) that indicates the UL transmission timing adjustment needed by a UE, as observed by the network. In general, a UE can determine a DL propagation delay from a TA MAC CE. Examples of TA MAC CEs include TA command MAC CE defined in 3GPP TS 38.321 section 6.1.3.4, absolute TA command MAC CE defined in 3GPP TS 38.321 section 6.1.3.4a, to-be-defined MAC CEs including an absolute TA with finer granularity than either of the above, etc.

(55) Even though the network transmits a TA MAC CE, it may not be successfully received by the intended UE. Since a MAC CE is not subject to retransmission protocols used on higher layers (e.g., RLC, PDCP), it can be lost during bad channel conditions without the network being aware that the UE did not receive the MAC CE, such as would be the case with higher-layer hybrid ARQ (HARQ) feedback from the UE. However, the network cannot simply rely on the UE's subsequent UL transmission to determine if the UE correctly received the TA MAC CE, such that a further TA MAC CE is not needed.

(56) In particular, a changed UL transmission timing can indicate either that the UE received and applied the TA MAC CE, or that the UE did not receive the TA MAC CE but adjusted its UL transmission timing due to UE movement or other propagation delay change. Similarly, a non-changed UL transmission timing can indicate either that the UE did not receive the TA MAC CE, or that the UE received and applied the TA MAC CE but also included an offsetting adjustment due to UE movement or other propagation delay change.

(57) In either of these scenarios, if the network transmits a second TA MAC CE without knowing whether the UE received and applied the first TA MAC CE, the network effectively loses knowledge of the UE-network propagation delay (PD). On the other hand, UE still knows the PD from received MAC CEs and/or from half of the timing advancement of the UL transmission in relation to the UE's received DL frame timing. In some cases, the network can pre-compensate the UE's PD but this requires that network is aware of the PD, which is not possible with the potential loss of the MAC CE.

(58) In some embodiments, a new timer `propDelayTimer` can be defined and can be configured for the UE's primary timing advance group (PTAG, i.e., the TAG for the UE's PCell) in one MAC entity to make sure that the TA MAC CE from the network is successfully received by the UE. This relies on the mechanism that the network periodically transmits the reference time as well as the PD compensation MAC CE.

(59) The value of `propDelayTimer` can be configured according to the periodicity of the reference time delivery, discussed above. Once the `propDelayTimer` timer value is configured by the network, the UE initiates or re-starts the timer with this value when a TA MAC CE is received at the UE. Upon expiration of the timer (e.g., without receiving a TA MAC CE), the UE can indicate in a new UL MAC CE that a previous DL TA MAC CE was not received.

(60) In some embodiments, an indicator (or "flag") can be added to the DL TA MAC CE. If a UE receives a DL TA MAC CE having the same indicator value as in the most recently received DL TA MAC CE, then the UE does not deliver the TA value in the MAC CE to the lower layer. On the other hand, if the UE receives a DL TA MAC CE with a different indicator value than in the most recently received DL TA MAC CE, then the UE delivers the TA value in the MAC CE to the lower layer. With this technique, the network can transmit two or more MAC CEs to the UE to ensure that at least one of them can be successfully received by the UE. FIG. 9 shows an exemplary implementation in which one of the reserved ("R") bits in an existing TA MAC CE has been repurposed into a flag ("F") bit according to these embodiments. The following exemplary text for 3GPP TS 38.321 (MAC specification) illustrates how a UE can interpret the exemplary MAC CE shown in FIG. 9.

(61) ***Begin exemplary text for 3GPP TS 38.321***

(62) 6.1.3.4b Absolute Timing Advance Command MAC CE with flag

(63) The Absolute Timing Advance Command MAC CE with flag is identified by MAC subheader with eLCID as specified in Table 6.2.1-1b.

(64) It has a fixed size and consists of two octets defined as follows (FIG. 6.1.3.4b-1):

(65) Timing Advance Command: This field indicates the index value TA used to control the amount of timing adjustment that the MAC entity has to apply in TS 38.213 [6]. The size of the field is 12 bits; R: Reserved bit, set to "0". F: The flag bit. If the same as the one before, UE ignores it. The initial value is zero

- (66) In other embodiments, the value of the propDelayTimer can be smaller than the value of the timeAlignmentTimer associated with the PTAG in the same MAC entity. This is because, upon the expiry of the timeAlignmentTimer, the UE does not receive the corresponding UL grant that can accommodate the UL MAC CE, which renders the propDelayTimer ineffective.
- (67) In some scenarios, a UE RRC connection to its serving gNB may fail completely, such as due to physical layer (PHY) failure, random access (RACH) failure, radio link control (RLC) layer failure, etc. In such scenarios, the UE can attempt to re-establish the RRC connection with another gNB in the manner currently defined in 3GPP specifications.
- (68) In some embodiments, the network can provide reference time information to UEs via a non-public network, e.g., SNPN or NPI-NPN. In such embodiments, the network can configure UEs to receive the reference time information via the SNPN or NPI-NPN. For example, the network can configure UEs to operate in SNPN access mode and/or configure an allowed CAG list that would provide the reference timing.
- (69) In some embodiments, a UE operating in a CAG or SNPN can send a UEAssistanceInformation message indicating that it prefers being provisioned with reference timing information. Upon receiving this indication, the serving NB can then distribute the reference timing information in the nearby cells in the same SNPN or within the same CAG.
- (70) In other embodiments that do not rely on such NPN-related techniques, when selecting a new cell after a failure event (or after selection of a new cell according to legacy methods), the UE determines whether the newly selected cell is capable of providing reference timing information. If not, the UE performs the same determination for the next suitable cell from a candidate list. This determination can involve the following operations: Check whether the broadcast cell status/reservation (e.g., in MIB or SIB1) includes a “reserved for reference time delivery” field. Such a field can be added to the fields defined in 3GPP TS 38.304 section 5.3. If the UE requires reference timing information, it should consider only cells that indicate delivery of such information. The UE connects to a new cell and then checks whether SIBS provides reference time information required by the UE. Alternately or in addition, the UE can include a request for reference time information in an RRCReestablishmentRequest message to the gNB serving the cell. The gNB can respond with an RRCReestablishment message that includes updated reference time information. On the other hand, if the gNB cannot provide the UE with reference time information, it may redirect (e.g., handover) the UE to another gNB that can provide such information.
- (71) In some embodiments, a UE can trigger and/or declare a RLF when the UE does not receive one or more expected reference time updates from the network. These embodiments can be used as alternatives to or in conjunction with embodiments that utilize a propDelayTimer, discussed above. For example, a UE can trigger RLF upon one or more consecutive expirations of the propDelayTimer without receiving a TA MAC CE. This could occur when the gNB is no longer able to provide reference time information. Upon declaring RLF, the UE can connect to another cell (e.g., via a reestablishment procedure) and inform the network via the new cell that the RLF cause was reference time synchronization failure (or more simply “sync failure”).
- (72) In other embodiments, the UE can trigger and/or declare RLF if a reference time update is not received for X duration (e.g., in ms) or during Y consecutive occasions when an update was expected.
- (73) The embodiments described above can be further illustrated with reference to FIGS. **10-11**, which depict exemplary methods (e.g., procedures) for a UE and a network node, respectively. Put differently, various features of the operations described below correspond to various embodiments described above. The exemplary methods shown in FIGS. **10-11** can be used cooperatively to provide various benefits, advantages, and/or solutions to problems, including those described herein. Although FIGS. **10-11** show specific blocks in particular orders, the operations corresponding to the blocks can be performed in different orders than shown and can be combined

and/or divided into blocks having different functionality than shown. Optional blocks and/or operations are indicated by dashed lines.

(74) In particular, FIG. **10** illustrates an exemplary method (e.g., procedure) for a user equipment (UE) to synchronize timing with a wireless network, according to various embodiments of the present disclosure. For example, the exemplary method shown in FIG. **10** can be implemented in a UE (e.g., wireless device, IoT device, etc.) such as described elsewhere herein.

(75) The exemplary method can include the operations of block **1040**, in which the UE can receive, from a first node serving a first cell in the wireless network, a first message that includes an absolute reference time and a corresponding reference event within downlink (DL) transmissions by the first node. The exemplary method can also include the operations of block **1050**, in which the UE can determine whether it received, from the first node, a timing adjustment value from which the UE can estimate a propagation delay of the DL transmissions from the first node to the UE.

(76) The exemplary method can also include the operations of block **1060**, in which the UE can perform one of various operations based on determining that it did not receive the timing adjustment value. In some embodiments, the UE can perform the operations of sub-block **1061**, where the UE can transmit, to the first node, an indication that the UE did not receive the timing adjustment value. In other embodiments, the UE can perform the operations of sub-block **1062**, where the UE can declare radio link failure (RLF) with respect to the first cell and establish a connection with a second cell that is capable of providing the information included in the first message.

(77) In some embodiments, the exemplary method can also include the operations of block **1080**, where the UE can receive, from the first node, a retransmission of the timing adjustment value after transmitting the indication (e.g., in sub-block **1061**).

(78) In some embodiments, the timing adjustment value is a timing advance (TA) command that indicates an amount of timing adjustment that the UE should apply to UL transmissions. In some of these embodiments, the determining operations of block **1050** can include the operations of sub-block **1051**, where the UE can determine whether the UE received a medium access control (MAC) control element (CE) comprising the timing adjustment value during a most recent first duration and/or a second duration after receiving the first message.

(79) In some of these embodiments, the exemplary method can also include the operations of blocks **1020-1030**, where the UE can receive, from the first node, a timer value associated with a timing advance group (TAG) that includes the first cell and initiate a timer based on the timer value upon receiving a previous timing adjustment value from the first node. In such embodiments, determining (e.g., in block **1050**) that the UE did not receive the timing adjustment value can be based on expiration of the timer initiated in this manner. Also, in some embodiments, transmitting (e.g., in block **1061**) the indication that the UE did not receive the timing adjustment value can be responsive to expiration of the timer. Furthermore, in some embodiments, declaring RLF (e.g., in block **1062**) can be based on the timer being initiated and expiring a plurality of consecutive times without receiving the timing adjustment value.

(80) In some embodiments, the exemplary method can also include the operations of blocks **1070**, where the UE can, based on determining that the UE did receive the timing adjustment value, selectively use the timing adjustment value for estimating the propagation delay of the DL transmissions. In some embodiments, the operations of block **1070** can include the operations of sub-block **1071**, where the UE can use the timing adjustment value only when the value of a flag, received with the TA command in the MAC CE, is different from a most recent value of the flag received with a most recent TA command.

(81) In some embodiments, the exemplary method can also include the operations of blocks **1010**, where the UE can send, to the first node, an indication of a UE preference for receiving reference time information from the wireless network. In such embodiments, receiving the first message

(e.g., in block **1040**) can be in response to the indication.

(82) In some embodiments, establishing a connection with a second cell in sub-block **1062** can include sub-operations **1062a-b**, where the UE can receive information broadcast by the second cell and determine that the broadcast information includes a field reserved for delivery of reference time information.

(83) In other embodiments, establishing a connection with a second cell in sub-block **1062** can include sub-operations **1062c-d**, where the UE can send, to a second node serving the second cell, a reestablishment request message comprising a request for reference time information; and receive, from the second node, a unicast reestablishment message comprising the requested reference time information.

(84) In some embodiments, the first node, the first cell, and the second cell can be part of a standalone non-public network (SNPN). In other embodiments, the first node can be part of a public network integrated non-public network (PNI-NPN), the UE can be part of a closed access group (CAG) associated with the PNI-NPN, and the first cell and the second cell are associated with the CAG.

(85) In addition, FIG. **11** illustrates an exemplary method (e.g., procedure) for a first node serving a first cell in a wireless network to synchronize timing with a UE, according to various embodiments of the present disclosure. For example, the exemplary method shown in FIG. **11** can be implemented in a network node (e.g., base station, eNB, gNB, etc., or component(s) thereof) such as described elsewhere herein.

(86) The exemplary method can include the operations of block **1130**, in which the first node can transmit, to the UE, a first message that includes an absolute reference time and a corresponding reference event within DL transmissions by the first node. The exemplary method can also include the operations of block **1150**, in which the first node can periodically transmit, to the UE, respective timing adjustment values from which the UE can estimate a propagation delay of the DL transmissions from the first node to the UE. The exemplary method can also include the operations of block **1170**, in which the first node can, based on determining that the UE did not receive one of the periodic transmissions of the respective timing adjustment values, retransmit the non-received timing adjustment value to the UE.

(87) In some embodiments, each timing adjustment value is a TA command that indicates an amount of timing adjustment that the UE should apply to UL transmissions. In addition, each timing adjustment value can be transmitted in a MAC CE.

(88) In some embodiments, the exemplary method can also include the operations of block **1120**, in which the first node can send, to the UE, a timer value associated with a TAG that includes the first cell. The timer value can be related to a period between successive transmissions of respective timing adjustment values. In some embodiments, the exemplary method can also include the operations of block **1160**, in which the first node can receive, from the UE, an indication that the UE did not receive the timing adjustment value during a duration corresponding to the timer value. The determination in block **1170** can be based on receiving this indication.

(89) In some embodiments, each timing adjustment value can be transmitted (e.g., in block **1150**) in a MAC CE together with a flag. In some embodiments, a first value of the flag can indicate that the timing adjustment value is different from a most recently transmitted timing adjustment value, and a second value of the flag can indicate that the timing adjustment value is the same as the most recently transmitted timing adjustment value.

(90) In some embodiments, the exemplary method can also include the operations of block **1110**, where the first node can receive, from the UE, an indication of the UE's preference for receiving reference time information from the wireless network. In such embodiments, the first message can be sent (e.g., in block **1130**) in response to this indication.

(91) In some embodiments, the first node and the first cell are part of an SNPN. In other embodiments, the first node can be part of a PNI-NPN, the UE can be part of a CAG associated

with the PNI-NPN, and the first cell is associated with the CAG. In such embodiments, the exemplary method can also include the operations of block **1140**, where the first node can transmit the first message in one or more further cells associated with the CAG.

(92) Although the subject matter described herein can be implemented in any appropriate type of system using any suitable components, the embodiments disclosed herein are described in relation to a wireless network, such as the example wireless network illustrated in FIG. **12**. For simplicity, the wireless network of FIG. **12** only depicts network **1206**, network nodes **1260** and **1260b**, and WDs **1210**, **1210b**, and **1210c**. In practice, a wireless network can further include any additional elements suitable to support communication between wireless devices or between a wireless device and another communication device, such as a landline telephone, a service provider, or any other network node or end device. Of the illustrated components, network node **1260** and wireless device (WD) **1210** are depicted with additional detail. The wireless network can provide communication and other types of services to one or more wireless devices to facilitate the wireless devices' access to and/or use of the services provided by, or via, the wireless network.

(93) The wireless network can comprise and/or interface with any type of communication, telecommunication, data, cellular, and/or radio network or other similar type of system. In some embodiments, the wireless network can be configured to operate according to specific standards or other types of predefined rules or procedures. Thus, particular embodiments of the wireless network can implement communication standards, such as Global System for Mobile Communications (GSM), Universal Mobile Telecommunications System (UMTS), Long Term Evolution (LTE), and/or other suitable 2G, 3G, 4G, or 5G standards; wireless local area network (WLAN) standards, such as the IEEE 802.11 standards; and/or any other appropriate wireless communication standard, such as the Worldwide Interoperability for Microwave Access (WiMax), Bluetooth, Z-Wave and/or ZigBee standards.

(94) Network **1206** can comprise one or more backhaul networks, core networks, IP networks, public switched telephone networks (PSTNs), packet data networks, optical networks, wide-area networks (WANs), local area networks (LANs), wireless local area networks (WLANs), wired networks, wireless networks, metropolitan area networks, and other networks to enable communication between devices.

(95) Network node **1260** and WD **1210** comprise various components described in more detail below. These components work together in order to provide network node and/or wireless device functionality, such as providing wireless connections in a wireless network. In different embodiments, the wireless network can comprise any number of wired or wireless networks, network nodes, base stations, controllers, wireless devices, relay stations, and/or any other components or systems that can facilitate or participate in the communication of data and/or signals whether via wired or wireless connections.

(96) Examples of network nodes include, but are not limited to, access points (APs) (e.g., radio access points), base stations (BSs) (e.g., radio base stations, Node Bs, evolved Node Bs (eNBs) and NR NodeBs (gNBs)). Base stations can be categorized based on the amount of coverage they provide (or, stated differently, their transmit power level) and can then also be referred to as femto base stations, pico base stations, micro base stations, or macro base stations. A base station can be a relay node or a relay donor node controlling a relay. A network node can also include one or more (or all) parts of a distributed radio base station such as centralized digital units and/or remote radio units (RRUs), sometimes referred to as Remote Radio Heads (RRHs). Such remote radio units may or may not be integrated with an antenna as an antenna integrated radio. Parts of a distributed radio base station can also be referred to as nodes in a distributed antenna system (DAS).

(97) Further examples of network nodes include multi-standard radio (MSR) equipment such as MSR BSs, network controllers such as radio network controllers (RNCs) or base station controllers (BSCs), base transceiver stations (BTSs), transmission points, transmission nodes, multi-cell/multicast coordination entities (MCEs), core network nodes (e.g., MSCs, MMEs), O&M

nodes, OSS nodes, SON nodes, positioning nodes (e.g., E-SMLCs), and/or MDTs. As another example, a network node can be a virtual network node as described in more detail below. More generally, however, network nodes can represent any suitable device (or group of devices) capable, configured, arranged, and/or operable to enable and/or provide a wireless device with access to the wireless network or to provide some service to a wireless device that has accessed the wireless network.

(98) In FIG. 12, network node **1260** includes processing circuitry **1270**, device readable medium **1280**, interface **1290**, auxiliary equipment **1284**, power source **1286**, power circuitry **1287**, and antenna **1262**. Although network node **1260** illustrated in the example wireless network of FIG. 12 can represent a device that includes the illustrated combination of hardware components, other embodiments can comprise network nodes with different combinations of components. It is to be understood that a network node comprises any suitable combination of hardware and/or software needed to perform the tasks, features, functions and methods and/or procedures disclosed herein. Moreover, while the components of network node **1260** are depicted as single boxes located within a larger box, or nested within multiple boxes, in practice, a network node can comprise multiple different physical components that make up a single illustrated component (e.g., device readable medium **1280** can comprise multiple separate hard drives as well as multiple RAM modules).

(99) Similarly, network node **1260** can be composed of multiple physically separate components (e.g., a NodeB component and a RNC component, or a BTS component and a BSC component, etc.), which can each have their own respective components. In certain scenarios in which network node **1260** comprises multiple separate components (e.g., BTS and BSC components), one or more of the separate components can be shared among several network nodes. For example, a single RNC can control multiple NodeB's. In such a scenario, each unique NodeB and RNC pair, can in some instances be considered a single separate network node. In some embodiments, network node **1260** can be configured to support multiple radio access technologies (RATs). In such embodiments, some components can be duplicated (e.g., separate device readable medium **1280** for the different RATs) and some components can be reused (e.g., the same antenna **1262** can be shared by the RATs). Network node **1260** can also include multiple sets of the various illustrated components for different wireless technologies integrated into network node **1260**, such as, for example, GSM, WCDMA, LTE, NR, WiFi, or Bluetooth wireless technologies. These wireless technologies can be integrated into the same or different chip or set of chips and other components within network node **1260**.

(100) Processing circuitry **1270** can be configured to perform any determining, calculating, or similar operations (e.g., certain obtaining operations) described herein as being provided by a network node. These operations performed by processing circuitry **1270** can include processing information obtained by processing circuitry **1270** by, for example, converting the obtained information into other information, comparing the obtained information or converted information to information stored in the network node, and/or performing one or more operations based on the obtained information or converted information, and as a result of said processing making a determination.

(101) Processing circuitry **1270** can comprise a combination of one or more of a microprocessor, controller, microcontroller, central processing unit, digital signal processor, application-specific integrated circuit, field programmable gate array, or any other suitable computing device, resource, or combination of hardware, software and/or encoded logic operable to provide various functionality of network node **1260**, either alone or in conjunction with other network node **1260** components (e.g., device readable medium **1280**). Such functionality can include any of the various wireless features, functions, or benefits discussed herein.

(102) For example, processing circuitry **1270** can execute instructions stored in device readable medium **1280** or in memory within processing circuitry **1270**. In some embodiments, processing circuitry **1270** can include a system on a chip (SOC). As a more specific example, instructions (also

referred to as a computer program product) stored in medium **1280** can include instructions that, when executed by processing circuitry **1270**, can configure network node **1260** to perform operations corresponding to various exemplary methods (e.g., procedures) described herein.

(103) In some embodiments, processing circuitry **1270** can include one or more of radio frequency (RF) transceiver circuitry **1272** and baseband processing circuitry **1274**. In some embodiments, radio frequency (RF) transceiver circuitry **1272** and baseband processing circuitry **1274** can be on separate chips (or sets of chips), boards, or units, such as radio units and digital units. In alternative embodiments, part or all of RF transceiver circuitry **1272** and baseband processing circuitry **1274** can be on the same chip or set of chips, boards, or units

(104) In certain embodiments, some or all of the functionality described herein as being provided by a network node, base station, eNB or other such network device can be performed by processing circuitry **1270** executing instructions stored on device readable medium **1280** or memory within processing circuitry **1270**. In alternative embodiments, some or all of the functionality can be provided by processing circuitry **1270** without executing instructions stored on a separate or discrete device readable medium, such as in a hard-wired manner. In any of those embodiments, whether executing instructions stored on a device readable storage medium or not, processing circuitry **1270** can be configured to perform the described functionality. The benefits provided by such functionality are not limited to processing circuitry **1270** alone or to other components of network node **1260** but are enjoyed by network node **1260** as a whole, and/or by end users and the wireless network generally.

(105) Device readable medium **1280** can comprise any form of volatile or non-volatile computer readable memory including, without limitation, persistent storage, solid-state memory, remotely mounted memory, magnetic media, optical media, random access memory (RAM), read-only memory (ROM), mass storage media (for example, a hard disk), removable storage media (for example, a flash drive, a Compact Disk (CD) or a Digital Video Disk (DVD)), and/or any other volatile or non-volatile, non-transitory device readable and/or computer-executable memory devices that store information, data, and/or instructions that can be used by processing circuitry **1270**. Device readable medium **1280** can store any suitable instructions, data or information, including a computer program, software, an application including one or more of logic, rules, code, tables, etc. and/or other instructions capable of being executed by processing circuitry **1270** and, utilized by network node **1260**. Device readable medium **1280** can be used to store any calculations made by processing circuitry **1270** and/or any data received via interface **1290**. In some embodiments, processing circuitry **1270** and device readable medium **1280** can be considered to be integrated.

(106) Interface **1290** is used in the wired or wireless communication of signaling and/or data between network node **1260**, network **1206**, and/or WDs **1210**. As illustrated, interface **1290** comprises port(s)/terminal(s) **1294** to send and receive data, for example to and from network **1206** over a wired connection. Interface **1290** also includes radio front end circuitry **1292** that can be coupled to, or in certain embodiments a part of, antenna **1262**. Radio front end circuitry **1292** comprises filters **1298** and amplifiers **1296**. Radio front end circuitry **1292** can be connected to antenna **1262** and processing circuitry **1270**. Radio front end circuitry can be configured to condition signals communicated between antenna **1262** and processing circuitry **1270**. Radio front end circuitry **1292** can receive digital data that is to be sent out to other network nodes or WDs via a wireless connection. Radio front end circuitry **1292** can convert the digital data into a radio signal having the appropriate channel and bandwidth parameters using a combination of filters **1298** and/or amplifiers **1296**. The radio signal can then be transmitted via antenna **1262**. Similarly, when receiving data, antenna **1262** can collect radio signals which are then converted into digital data by radio front end circuitry **1292**. The digital data can be passed to processing circuitry **1270**. In other embodiments, the interface can comprise different components and/or different combinations of components.

(107) In certain alternative embodiments, network node **1260** may not include separate radio front end circuitry **1292**, instead, processing circuitry **1270** can comprise radio front end circuitry and can be connected to antenna **1262** without separate radio front end circuitry **1292**. Similarly, in some embodiments, all or some of RF transceiver circuitry **1272** can be considered a part of interface **1290**. In still other embodiments, interface **1290** can include one or more ports or terminals **1294**, radio front end circuitry **1292**, and RF transceiver circuitry **1272**, as part of a radio unit (not shown), and interface **1290** can communicate with baseband processing circuitry **1274**, which is part of a digital unit (not shown).

(108) Antenna **1262** can include one or more antennas, or antenna arrays, configured to send and/or receive wireless signals. Antenna **1262** can be coupled to radio front end circuitry **1290** and can be any type of antenna capable of transmitting and receiving data and/or signals wirelessly. In some embodiments, antenna **1262** can comprise one or more omni-directional, sector or panel antennas operable to transmit/receive radio signals between, for example, 2 GHz and 66 GHz. An omni-directional antenna can be used to transmit/receive radio signals in any direction, a sector antenna can be used to transmit/receive radio signals from devices within a particular area, and a panel antenna can be a line of sight antenna used to transmit/receive radio signals in a relatively straight line. In some instances, the use of more than one antenna can be referred to as MIMO. In certain embodiments, antenna **1262** can be separate from network node **1260** and can be connectable to network node **1260** through an interface or port.

(109) Antenna **1262**, interface **1290**, and/or processing circuitry **1270** can be configured to perform any receiving operations and/or certain obtaining operations described herein as being performed by a network node. Any information, data and/or signals can be received from a wireless device, another network node and/or any other network equipment. Similarly, antenna **1262**, interface **1290**, and/or processing circuitry **1270** can be configured to perform any transmitting operations described herein as being performed by a network node. Any information, data and/or signals can be transmitted to a wireless device, another network node and/or any other network equipment.

(110) Power circuitry **1287** can comprise, or be coupled to, power management circuitry and can be configured to supply the components of network node **1260** with power for performing the functionality described herein. Power circuitry **1287** can receive power from power source **1286**. Power source **1286** and/or power circuitry **1287** can be configured to provide power to the various components of network node **1260** in a form suitable for the respective components (e.g., at a voltage and current level needed for each respective component). Power source **1286** can either be included in, or external to, power circuitry **1287** and/or network node **1260**. For example, network node **1260** can be connectable to an external power source (e.g., an electricity outlet) via an input circuitry or interface such as an electrical cable, whereby the external power source supplies power to power circuitry **1287**. As a further example, power source **1286** can comprise a source of power in the form of a battery or battery pack which is connected to, or integrated in, power circuitry **1287**. The battery can provide backup power should the external power source fail. Other types of power sources, such as photovoltaic devices, can also be used.

(111) Alternative embodiments of network node **1260** can include additional components beyond those shown in FIG. 12 that can be responsible for providing certain aspects of the network node's functionality, including any of the functionality described herein and/or any functionality necessary to support the subject matter described herein. For example, network node **1260** can include user interface equipment to allow and/or facilitate input of information into network node **1260** and to allow and/or facilitate output of information from network node **1260**. This can allow and/or facilitate a user to perform diagnostic, maintenance, repair, and other administrative functions for network node **1260**.

(112) In some embodiments, a wireless device (WD, e.g., WD **1210**) can be configured to transmit and/or receive information without direct human interaction. For instance, a WD can be designed to transmit information to a network on a predetermined schedule, when triggered by an internal or

external event, or in response to requests from the network. Examples of a WD include, but are not limited to, smart phones, mobile phones, cell phones, voice over IP (VoIP) phones, wireless local loop phones, desktop computers, personal digital assistants (PDAs), wireless cameras, gaming consoles or devices, music storage devices, playback appliances, wearable devices, wireless endpoints, mobile stations, tablets, laptops, laptop-embedded equipment (LEE), laptop-mounted equipment (LME), smart devices, wireless customer-premise equipment (CPE), mobile-type communication (MTC) devices, Internet-of-Things (IoT) devices, vehicle-mounted wireless terminal devices, etc.

(113) A WD can support device-to-device (D2D) communication, for example by implementing a 3GPP standard for sidelink communication, vehicle-to-vehicle (V2V), vehicle-to-infrastructure (V2I), vehicle-to-everything (V2X) and can in this case be referred to as a D2D communication device. As yet another specific example, in an Internet of Things (IoT) scenario, a WD can represent a machine or other device that performs monitoring and/or measurements and transmits the results of such monitoring and/or measurements to another WD and/or a network node. The WD can in this case be a machine-to-machine (M2M) device, which can in a 3GPP context be referred to as an MTC device. As one particular example, the WD can be a UE implementing the 3GPP narrow band internet of things (NB-IoT) standard. Particular examples of such machines or devices are sensors, metering devices such as power meters, industrial machinery, or home or personal appliances (e.g., refrigerators, televisions, etc.) personal wearables (e.g., watches, fitness trackers, etc.). In other scenarios, a WD can represent a vehicle or other equipment that is capable of monitoring and/or reporting on its operational status or other functions associated with its operation. A WD as described above can represent the endpoint of a wireless connection, in which case the device can be referred to as a wireless terminal. Furthermore, a WD as described above can be mobile, in which case it can also be referred to as a mobile device or a mobile terminal.

(114) As illustrated, wireless device **1210** includes antenna **1211**, interface **1214**, processing circuitry **1220**, device readable medium **1230**, user interface equipment **1232**, auxiliary equipment **1234**, power source **1236** and power circuitry **1237**. WD **1210** can include multiple sets of one or more of the illustrated components for different wireless technologies supported by WD **1210**, such as, for example, GSM, WCDMA, LTE, NR, WiFi, WiMAX, or Bluetooth wireless technologies, just to mention a few. These wireless technologies can be integrated into the same or different chips or set of chips as other components within WD **1210**.

(115) Antenna **1211** can include one or more antennas or antenna arrays, configured to send and/or receive wireless signals, and is connected to interface **1214**. In certain alternative embodiments, antenna **1211** can be separate from WD **1210** and be connectable to WD **1210** through an interface or port. Antenna **1211**, interface **1214**, and/or processing circuitry **1220** can be configured to perform any receiving or transmitting operations described herein as being performed by a WD. Any information, data and/or signals can be received from a network node and/or another WD. In some embodiments, radio front end circuitry and/or antenna **1211** can be considered an interface.

(116) As illustrated, interface **1214** comprises radio front end circuitry **1212** and antenna **1211**. Radio front end circuitry **1212** comprise one or more filters **1218** and amplifiers **1216**. Radio front end circuitry **1214** is connected to antenna **1211** and processing circuitry **1220** and can be configured to condition signals communicated between antenna **1211** and processing circuitry **1220**. Radio front end circuitry **1212** can be coupled to or a part of antenna **1211**. In some embodiments, WD **1210** may not include separate radio front end circuitry **1212**; rather, processing circuitry **1220** can comprise radio front end circuitry and can be connected to antenna **1211**. Similarly, in some embodiments, some or all of RF transceiver circuitry **1222** can be considered a part of interface **1214**. Radio front end circuitry **1212** can receive digital data that is to be sent out to other network nodes or WDs via a wireless connection. Radio front end circuitry **1212** can convert the digital data into a radio signal having the appropriate channel and bandwidth parameters using a combination of filters **1218** and/or amplifiers **1216**. The radio signal can then be

transmitted via antenna **1211**. Similarly, when receiving data, antenna **1211** can collect radio signals which are then converted into digital data by radio front end circuitry **1212**. The digital data can be passed to processing circuitry **1220**. In other embodiments, the interface can comprise different components and/or different combinations of components.

(117) Processing circuitry **1220** can comprise a combination of one or more of a microprocessor, controller, microcontroller, central processing unit, digital signal processor, application-specific integrated circuit, field programmable gate array, or any other suitable computing device, resource, or combination of hardware, software, and/or encoded logic operable to provide WD **1210** functionality either alone or in combination with other WD **1210** components, such as device readable medium **1230**. Such functionality can include any of the various wireless features or benefits discussed herein.

(118) For example, processing circuitry **1220** can execute instructions stored in device readable medium **1230** or in memory within processing circuitry **1220** to provide the functionality disclosed herein. More specifically, instructions (also referred to as a computer program product) stored in medium **1230** can include instructions that, when executed by processor **1220**, can configure wireless device **1210** to perform operations corresponding to various exemplary methods (e.g., procedures) described herein.

(119) As illustrated, processing circuitry **1220** includes one or more of RF transceiver circuitry **1222**, baseband processing circuitry **1224**, and application processing circuitry **1226**. In other embodiments, the processing circuitry can comprise different components and/or different combinations of components. In certain embodiments processing circuitry **1220** of WD **1210** can comprise a SOC. In some embodiments, RF transceiver circuitry **1222**, baseband processing circuitry **1224**, and application processing circuitry **1226** can be on separate chips or sets of chips. In alternative embodiments, part or all of baseband processing circuitry **1224** and application processing circuitry **1226** can be combined into one chip or set of chips, and RF transceiver circuitry **1222** can be on a separate chip or set of chips. In still alternative embodiments, part or all of RF transceiver circuitry **1222** and baseband processing circuitry **1224** can be on the same chip or set of chips, and application processing circuitry **1226** can be on a separate chip or set of chips. In yet other alternative embodiments, part or all of RF transceiver circuitry **1222**, baseband processing circuitry **1224**, and application processing circuitry **1226** can be combined in the same chip or set of chips. In some embodiments, RF transceiver circuitry **1222** can be a part of interface **1214**. RF transceiver circuitry **1222** can condition RF signals for processing circuitry **1220**.

(120) In certain embodiments, some or all of the functionality described herein as being performed by a WD can be provided by processing circuitry **1220** executing instructions stored on device readable medium **1230**, which in certain embodiments can be a computer-readable storage medium. In alternative embodiments, some or all of the functionality can be provided by processing circuitry **1220** without executing instructions stored on a separate or discrete device readable storage medium, such as in a hard-wired manner. In any of those particular embodiments, whether executing instructions stored on a device readable storage medium or not, processing circuitry **1220** can be configured to perform the described functionality. The benefits provided by such functionality are not limited to processing circuitry **1220** alone or to other components of WD **1210**, but are enjoyed by WD **1210** as a whole, and/or by end users and the wireless network generally.

(121) Processing circuitry **1220** can be configured to perform any determining, calculating, or similar operations (e.g., certain obtaining operations) described herein as being performed by a WD. These operations, as performed by processing circuitry **1220**, can include processing information obtained by processing circuitry **1220** by, for example, converting the obtained information into other information, comparing the obtained information or converted information to information stored by WD **1210**, and/or performing one or more operations based on the obtained information or converted information, and as a result of said processing making a

determination.

(122) Device readable medium **1230** can be operable to store a computer program, software, an application including one or more of logic, rules, code, tables, etc. and/or other instructions capable of being executed by processing circuitry **1220**. Device readable medium **1230** can include computer memory (e.g., Random Access Memory (RAM) or Read Only Memory (ROM)), mass storage media (e.g., a hard disk), removable storage media (e.g., a Compact Disk (CD) or a Digital Video Disk (DVD)), and/or any other volatile or non-volatile, non-transitory device readable and/or computer executable memory devices that store information, data, and/or instructions that can be used by processing circuitry **1220**. In some embodiments, processing circuitry **1220** and device readable medium **1230** can be considered to be integrated.

(123) User interface equipment **1232** can include components that allow and/or facilitate a human user to interact with WD **1210**. Such interaction can be of many forms, such as visual, audial, tactile, etc. User interface equipment **1232** can be operable to produce output to the user and to allow and/or facilitate the user to provide input to WD **1210**. The type of interaction can vary depending on the type of user interface equipment **1232** installed in WD **1210**. For example, if WD **1210** is a smart phone, the interaction can be via a touch screen; if WD **1210** is a smart meter, the interaction can be through a screen that provides usage (e.g., the number of gallons used) or a speaker that provides an audible alert (e.g., if smoke is detected). User interface equipment **1232** can include input interfaces, devices and circuits, and output interfaces, devices and circuits. User interface equipment **1232** can be configured to allow and/or facilitate input of information into WD **1210** and is connected to processing circuitry **1220** to allow and/or facilitate processing circuitry **1220** to process the input information. User interface equipment **1232** can include, for example, a microphone, a proximity or other sensor, keys/buttons, a touch display, one or more cameras, a USB port, or other input circuitry. User interface equipment **1232** is also configured to allow and/or facilitate output of information from WD **1210**, and to allow and/or facilitate processing circuitry **1220** to output information from WD **1210**. User interface equipment **1232** can include, for example, a speaker, a display, vibrating circuitry, a USB port, a headphone interface, or other output circuitry. Using one or more input and output interfaces, devices, and circuits, of user interface equipment **1232**, WD **1210** can communicate with end users and/or the wireless network and allow and/or facilitate them to benefit from the functionality described herein.

(124) Auxiliary equipment **1234** is operable to provide more specific functionality which may not be generally performed by WDs. This can comprise specialized sensors for doing measurements for various purposes, interfaces for additional types of communication such as wired communications etc. The inclusion and type of components of auxiliary equipment **1234** can vary depending on the embodiment and/or scenario.

(125) Power source **1236** can, in some embodiments, be in the form of a battery or battery pack. Other types of power sources, such as an external power source (e.g., an electricity outlet), photovoltaic devices or power cells, can also be used. WD **1210** can further comprise power circuitry **1237** for delivering power from power source **1236** to the various parts of WD **1210** which need power from power source **1236** to carry out any functionality described or indicated herein. Power circuitry **1237** can in certain embodiments comprise power management circuitry. Power circuitry **1237** can additionally or alternatively be operable to receive power from an external power source; in which case WD **1210** can be connectable to the external power source (such as an electricity outlet) via input circuitry or an interface such as an electrical power cable. Power circuitry **1237** can also in certain embodiments be operable to deliver power from an external power source to power source **1236**. This can be, for example, for the charging of power source **1236**. Power circuitry **1237** can perform any converting or other modification to the power from power source **1236** to make it suitable for supply to the respective components of WD **1210**.

(126) FIG. **13** illustrates one embodiment of a UE in accordance with various aspects described herein. As used herein, a user equipment or UE may not necessarily have a user in the sense of a

human user who owns and/or operates the relevant device. Instead, a UE can represent a device that is intended for sale to, or operation by, a human user but which may not, or which may not initially, be associated with a specific human user (e.g., a smart sprinkler controller). Alternatively, a UE can represent a device that is not intended for sale to, or operation by, an end user but which can be associated with or operated for the benefit of a user (e.g., a smart power meter). UE **13200** can be any UE identified by the 3rd Generation Partnership Project (3GPP), including a NB-IoT UE, a machine type communication (MTC) UE, and/or an enhanced MTC (eMTC) UE. UE **1300**, as illustrated in FIG. **13**, is one example of a WD configured for communication in accordance with one or more communication standards promulgated by the 3rd Generation Partnership Project (3GPP), such as 3GPP's GSM, UMTS, LTE, and/or 5G standards. As mentioned previously, the term WD and UE can be used interchangeable. Accordingly, although FIG. **13** is a UE, the components discussed herein are equally applicable to a WD, and vice-versa.

(127) In FIG. **13**, UE **1300** includes processing circuitry **1301** that is operatively coupled to input/output interface **1305**, radio frequency (RF) interface **1309**, network connection interface **1311**, memory **1315** including random access memory (RAM) **1317**, read-only memory (ROM) **1319**, and storage medium **1321** or the like, communication subsystem **1331**, power source **1333**, and/or any other component, or any combination thereof. Storage medium **1321** includes operating system **1323**, application program **1325**, and data **1327**. In other embodiments, storage medium **1321** can include other similar types of information. Certain UEs can utilize all of the components shown in FIG. **13**, or only a subset of the components. The level of integration between the components can vary from one UE to another UE. Further, certain UEs can contain multiple instances of a component, such as multiple processors, memories, transceivers, transmitters, receivers, etc.

(128) In FIG. **13**, processing circuitry **1301** can be configured to process computer instructions and data. Processing circuitry **1301** can be configured to implement any sequential state machine operative to execute machine instructions stored as machine-readable computer programs in the memory, such as one or more hardware-implemented state machines (e.g., in discrete logic, FPGA, ASIC, etc.); programmable logic together with appropriate firmware; one or more stored program, general-purpose processors, such as a microprocessor or Digital Signal Processor (DSP), together with appropriate software; or any combination of the above. For example, the processing circuitry **1301** can include two central processing units (CPUs). Data can be information in a form suitable for use by a computer.

(129) In the depicted embodiment, input/output interface **1305** can be configured to provide a communication interface to an input device, output device, or input and output device. UE **1300** can be configured to use an output device via input/output interface **1305**. An output device can use the same type of interface port as an input device. For example, a USB port can be used to provide input to and output from UE **1300**. The output device can be a speaker, a sound card, a video card, a display, a monitor, a printer, an actuator, an emitter, a smartcard, another output device, or any combination thereof. UE **1300** can be configured to use an input device via input/output interface **1305** to allow and/or facilitate a user to capture information into UE **1300**. The input device can include a touch-sensitive or presence-sensitive display, a camera (e.g., a digital camera, a digital video camera, a web camera, etc.), a microphone, a sensor, a mouse, a trackball, a directional pad, a trackpad, a scroll wheel, a smartcard, and the like. The presence-sensitive display can include a capacitive or resistive touch sensor to sense input from a user. A sensor can be, for instance, an accelerometer, a gyroscope, a tilt sensor, a force sensor, a magnetometer, an optical sensor, a proximity sensor, another like sensor, or any combination thereof. For example, the input device can be an accelerometer, a magnetometer, a digital camera, a microphone, and an optical sensor.

(130) In FIG. **13**, RF interface **1309** can be configured to provide a communication interface to RF components such as a transmitter, a receiver, and an antenna. Network connection interface **1311** can be configured to provide a communication interface to network **1343a**. Network **1343a** can

encompass wired and/or wireless networks such as a local-area network (LAN), a wide-area network (WAN), a computer network, a wireless network, a telecommunications network, another like network or any combination thereof. For example, network **1343a** can comprise a Wi-Fi network. Network connection interface **1311** can be configured to include a receiver and a transmitter interface used to communicate with one or more other devices over a communication network according to one or more communication protocols, such as Ethernet, TCP/IP, SONET, ATM, or the like. Network connection interface **1311** can implement receiver and transmitter functionality appropriate to the communication network links (e.g., optical, electrical, and the like). The transmitter and receiver functions can share circuit components, software or firmware, or alternatively can be implemented separately.

(131) RAM **1317** can be configured to interface via bus **1302** to processing circuitry **1301** to provide storage or caching of data or computer instructions during the execution of software programs such as the operating system, application programs, and device drivers. ROM **1319** can be configured to provide computer instructions or data to processing circuitry **1301**. For example, ROM **1319** can be configured to store invariant low-level system code or data for basic system functions such as basic input and output (I/O), startup, or reception of keystrokes from a keyboard that are stored in a non-volatile memory. Storage medium **1321** can be configured to include memory such as RAM, ROM, programmable read-only memory (PROM), erasable programmable read-only memory (EPROM), electrically erasable programmable read-only memory (EEPROM), magnetic disks, optical disks, floppy disks, hard disks, removable cartridges, or flash drives.

(132) In one example, storage medium **1321** can be configured to include operating system **1323**; application program **1325** such as a web browser application, a widget or gadget engine or another application; and data file **1327**. Storage medium **1321** can store, for use by UE **1300**, any of a variety of various operating systems or combinations of operating systems. For example, application program **1325** can include executable program instructions (also referred to as a computer program product) that, when executed by processor **1301**, can configure UE **1300** to perform operations corresponding to various exemplary methods (e.g., procedures) described herein.

(133) Storage medium **1321** can be configured to include a number of physical drive units, such as redundant array of independent disks (RAID), floppy disk drive, flash memory, USB flash drive, external hard disk drive, thumb drive, pen drive, key drive, high-density digital versatile disc (HD-DVD) optical disc drive, internal hard disk drive, Blu-Ray optical disc drive, holographic digital data storage (HDDS) optical disc drive, external mini-dual in-line memory module (DIMM), synchronous dynamic random access memory (SDRAM), external micro-DIMM SDRAM, smartcard memory such as a subscriber identity module or a removable user identity (SIM/RUIM) module, other memory, or any combination thereof. Storage medium **1321** can allow and/or facilitate UE **1300** to access computer-executable instructions, application programs or the like, stored on transitory or non-transitory memory media, to off-load data, or to upload data. An article of manufacture, such as one utilizing a communication system can be tangibly embodied in storage medium **1321**, which can comprise a device readable medium.

(134) In FIG. **13**, processing circuitry **1301** can be configured to communicate with network **1343b** using communication subsystem **1331**. Network **1343a** and network **1343b** can be the same network or networks or different network or networks. Communication subsystem **1331** can be configured to include one or more transceivers used to communicate with network **1343b**. For example, communication subsystem **1331** can be configured to include one or more transceivers used to communicate with one or more remote transceivers of another device capable of wireless communication such as another WD, UE, or base station of a radio access network (RAN) according to one or more communication protocols, such as IEEE 802.13, CDMA, WCDMA, GSM, LTE, UTRAN, WiMax, or the like. Each transceiver can include transmitter **1333** and/or receiver **1335** to implement transmitter or receiver functionality, respectively, appropriate to the

RAN links (e.g., frequency allocations and the like). Further, transmitter **1333** and receiver **1335** of each transceiver can share circuit components, software or firmware, or alternatively can be implemented separately.

(135) In the illustrated embodiment, the communication functions of communication subsystem **1331** can include data communication, voice communication, multimedia communication, short-range communications such as Bluetooth, near-field communication, location-based communication such as the use of the global positioning system (GPS) to determine a location, another like communication function, or any combination thereof. For example, communication subsystem **1331** can include cellular communication, Wi-Fi communication, Bluetooth communication, and GPS communication. Network **1343b** can encompass wired and/or wireless networks such as a local-area network (LAN), a wide-area network (WAN), a computer network, a wireless network, a telecommunications network, another like network or any combination thereof. For example, network **1343b** can be a cellular network, a Wi-Fi network, and/or a near-field network. Power source **1313** can be configured to provide alternating current (AC) or direct current (DC) power to components of UE **1300**.

(136) The features, benefits and/or functions described herein can be implemented in one of the components of UE **1300** or partitioned across multiple components of UE **1300**. Further, the features, benefits, and/or functions described herein can be implemented in any combination of hardware, software or firmware. In one example, communication subsystem **1331** can be configured to include any of the components described herein. Further, processing circuitry **1301** can be configured to communicate with any of such components over bus **1302**. In another example, any of such components can be represented by program instructions stored in memory that when executed by processing circuitry **1301** perform the corresponding functions described herein. In another example, the functionality of any of such components can be partitioned between processing circuitry **1301** and communication subsystem **1331**. In another example, the non-computationally intensive functions of any of such components can be implemented in software or firmware and the computationally intensive functions can be implemented in hardware.

(137) FIG. **14** is a schematic block diagram illustrating a virtualization environment **1400** in which functions implemented by some embodiments can be virtualized. In the present context, virtualizing means creating virtual versions of apparatuses or devices which can include virtualizing hardware platforms, storage devices and networking resources. As used herein, virtualization can be applied to a node (e.g., a virtualized base station or a virtualized radio access node) or to a device (e.g., a UE, a wireless device or any other type of communication device) or components thereof and relates to an implementation in which at least a portion of the functionality is implemented as one or more virtual components (e.g., via one or more applications, components, functions, virtual machines or containers executing on one or more physical processing nodes in one or more networks).

(138) In some embodiments, some or all of the functions described herein can be implemented as virtual components executed by one or more virtual machines implemented in one or more virtual environments **1400** hosted by one or more of hardware nodes **1430**. Further, in embodiments in which the virtual node is not a radio access node or does not require radio connectivity (e.g., a core network node), then the network node can be entirely virtualized.

(139) The functions can be implemented by one or more applications **1420** (which can alternatively be called software instances, virtual appliances, network functions, virtual nodes, virtual network functions, etc.) operative to implement some of the features, functions, and/or benefits of some of the embodiments disclosed herein. Applications **1420** are run in virtualization environment **1400** which provides hardware **1430** comprising processing circuitry **1460** and memory **1490**. Memory **1490** contains instructions **1495** executable by processing circuitry **1460** whereby application **1420** is operative to provide one or more of the features, benefits, and/or functions disclosed herein.

(140) Virtualization environment **1400** can include general-purpose or special-purpose network

hardware devices (or nodes) **1430** comprising a set of one or more processors or processing circuitry **1460**, which can be commercial off-the-shelf (COTS) processors, dedicated Application Specific Integrated Circuits (ASICs), or any other type of processing circuitry including digital or analog hardware components or special purpose processors. Each hardware device can comprise memory **1490-1** which can be non-persistent memory for temporarily storing instructions **1495** or software executed by processing circuitry **1460**. For example, instructions **1495** can include program instructions (also referred to as a computer program product) that, when executed by processing circuitry **1460**, can configure hardware node **1420** to perform operations corresponding to various exemplary methods (e.g., procedures) described herein. Such operations can also be attributed to virtual node(s) **1420** that is/are hosted by hardware node **1430**.

(141) Each hardware device can comprise one or more network interface controllers (NICs) **1470**, also known as network interface cards, which include physical network interface **1480**. Each hardware device can also include non-transitory, persistent, machine-readable storage media **1490-2** having stored therein software **1495** and/or instructions executable by processing circuitry **1460**. Software **1495** can include any type of software including software for instantiating one or more virtualization layers **1450** (also referred to as hypervisors), software to execute virtual machines **1440** as well as software allowing it to execute functions, features and/or benefits described in relation with some embodiments described herein.

(142) Virtual machines **1440**, comprise virtual processing, virtual memory, virtual networking or interface and virtual storage, and can be run by a corresponding virtualization layer **1450** or hypervisor. Different embodiments of the instance of virtual appliance **1420** can be implemented on one or more of virtual machines **1440**, and the implementations can be made in different ways.

(143) During operation, processing circuitry **1460** executes software **1495** to instantiate the hypervisor or virtualization layer **1450**, which can sometimes be referred to as a virtual machine monitor (VMM). Virtualization layer **1450** can present a virtual operating platform that appears like networking hardware to virtual machine **1440**.

(144) As shown in FIG. 14, hardware **1430** can be a standalone network node with generic or specific components. Hardware **1430** can comprise antenna **14225** and can implement some functions via virtualization. Alternatively, hardware **1430** can be part of a larger cluster of hardware (e.g., such as in a data center or customer premise equipment (CPE)) where many hardware nodes work together and are managed via management and orchestration (MANO) **14100**, which, among others, oversees lifecycle management of applications **1420**.

(145) Virtualization of the hardware is in some contexts referred to as network function virtualization (NFV). NFV can be used to consolidate many network equipment types onto industry standard high volume server hardware, physical switches, and physical storage, which can be located in data centers, and customer premise equipment.

(146) In the context of NFV, virtual machine **1440** can be a software implementation of a physical machine that runs programs as if they were executing on a physical, non-virtualized machine. Each of virtual machines **1440**, and that part of hardware **1430** that executes that virtual machine, be it hardware dedicated to that virtual machine and/or hardware shared by that virtual machine with others of the virtual machines **1440**, forms a separate virtual network elements (VNE).

(147) Still in the context of NFV, Virtual Network Function (VNF) is responsible for handling specific network functions that run in one or more virtual machines **1440** on top of hardware networking infrastructure **1430** and corresponds to application **1420** in FIG. 14.

(148) In some embodiments, one or more radio units **14200** that each include one or more transmitters **14220** and one or more receivers **14210** can be coupled to one or more antennas **14225**. Radio units **14200** can communicate directly with hardware nodes **1430** via one or more appropriate network interfaces and can be used in combination with the virtual components to provide a virtual node with radio capabilities, such as a radio access node or a base station. Nodes arranged in this manner can also communicate with one or more UEs, such as described elsewhere

herein.

(149) In some embodiments, some signaling can be performed via control system **14230**, which can alternatively be used for communication between the hardware nodes **1430** and radio units **14200**.

(150) With reference to FIG. **15**, in accordance with an embodiment, a communication system includes telecommunication network **1510**, such as a 3GPP-type cellular network, which comprises access network **1511**, such as a radio access network, and core network **1514**. Access network **1511** comprises a plurality of base stations **1512a**, **1512b**, **1512c**, such as NBs, eNBs, gNBs or other types of wireless access points, each defining a corresponding coverage area **1513a**, **1513b**, **1513c**. Each base station **1512a**, **1512b**, **1512c** is connectable to core network **1514** over a wired or wireless connection **1515**. A first UE **1591** located in coverage area **1513c** can be configured to wirelessly connect to, or be paged by, the corresponding base station **1512c**. A second UE **1592** in coverage area **1513a** is wirelessly connectable to the corresponding base station **1512a**. While a plurality of UEs **1591**, **1592** are illustrated in this example, the disclosed embodiments are equally applicable to a situation where a sole UE is in the coverage area or where a sole UE is connecting to the base stations in the coverage area.

(151) Telecommunication network **1510** is itself connected to host computer **1530**, which can be embodied in the hardware and/or software of a standalone server, a cloud-implemented server, a distributed server or as processing resources in a server farm. Host computer **1530** can be under the ownership or control of a service provider or can be operated by the service provider or on behalf of the service provider. Connections **1521** and **1522** between telecommunication network **1510** and host computer **1530** can extend directly from core network **1514** to host computer **1530** to or can go via an optional intermediate network **1520**. Intermediate network **1520** can be one of, or a combination of more than one of, a public, private or hosted network; intermediate network **1520**, if any, can be a backbone network or the Internet; in particular, intermediate network **1520** can comprise two or more sub-networks (not shown).

(152) The communication system of FIG. **15** as a whole enables connectivity between the connected UEs **1591**, **1592** and host computer **1530**. The connectivity can be described as an over-the-top (OTT) connection **1550**. Host computer **1530** and the connected UEs **1591**, **1592** are configured to communicate data and/or signaling via OTT connection **1550**, using access network **1511**, core network **1514**, any intermediate network **1520** and possible further infrastructure (not shown) as intermediaries. OTT connection **1550** can be transparent in the sense that the participating communication devices through which OTT connection **1550** passes are unaware of routing of uplink and downlink communications. For example, base station **1512** may not or need not be informed about the past routing of an incoming downlink communication with data originating from host computer **1530** to be forwarded (e.g., handed over) to a connected UE **1591**. Similarly, base station **1512** need not be aware of the future routing of an outgoing uplink communication originating from the UE **1591** towards the host computer **1530**.

(153) Example implementations, in accordance with an embodiment, of the UE, base station and host computer discussed in the preceding paragraphs will now be described with reference to FIG. **16**. In communication system **1600**, host computer **1610** comprises hardware **1615** including communication interface **1616** configured to set up and maintain a wired or wireless connection with an interface of a different communication device of communication system **1600**. Host computer **1610** further comprises processing circuitry **1618**, which can have storage and/or processing capabilities. In particular, processing circuitry **1618** can comprise one or more programmable processors, application-specific integrated circuits, field programmable gate arrays or combinations of these (not shown) adapted to execute instructions. Host computer **1610** further comprises software **1611**, which is stored in or accessible by host computer **1610** and executable by processing circuitry **1618**. Software **1611** includes host application **1612**. Host application **1612** can be operable to provide a service to a remote user, such as UE **1630** connecting via OTT connection

1650 terminating at UE **1630** and host computer **1610**. In providing the service to the remote user, host application **1612** can provide user data which is transmitted using OTT connection **1650**.

(154) Communication system **1600** can also include base station **1620** provided in a telecommunication system and comprising hardware **1625** enabling it to communicate with host computer **1610** and with UE **1630**. Hardware **1625** can include communication interface **1626** for setting up and maintaining a wired or wireless connection with an interface of a different communication device of communication system **1600**, as well as radio interface **1627** for setting up and maintaining at least wireless connection **1670** with UE **1630** located in a coverage area (not shown in FIG. **16**) served by base station **1620**. Communication interface **1626** can be configured to facilitate connection **1660** to host computer **1610**. Connection **1660** can be direct, or it can pass through a core network (not shown in FIG. **16**) of the telecommunication system and/or through one or more intermediate networks outside the telecommunication system. In the embodiment shown, hardware **1625** of base station **1620** can also include processing circuitry **1628**, which can comprise one or more programmable processors, application-specific integrated circuits, field programmable gate arrays or combinations of these (not shown) adapted to execute instructions. (155) Base station **1620** also includes software **1621** stored internally or accessible via an external connection. For example, software **1621** can include program instructions (also referred to as a computer program product) that, when executed by processing circuitry **1628**, can configure base station **1620** to perform operations corresponding to various exemplary methods (e.g., procedures) described herein.

(156) Communication system **1600** can also include UE **1630** already referred to, whose hardware **1635** can include radio interface **1637** configured to set up and maintain wireless connection **1670** with a base station serving a coverage area in which UE **1630** is currently located. Hardware **1635** of UE **1630** can also include processing circuitry **1638**, which can comprise one or more programmable processors, application-specific integrated circuits, field programmable gate arrays or combinations of these (not shown) adapted to execute instructions.

(157) UE **1630** also includes software **1631**, which is stored in or accessible by UE **1630** and executable by processing circuitry **1638**. Software **1631** includes client application **1632**. Client application **1632** can be operable to provide a service to a human or non-human user via UE **1630**, with the support of host computer **1610**. In host computer **1610**, an executing host application **1612** can communicate with the executing client application **1632** via OTT connection **1650** terminating at UE **1630** and host computer **1610**. In providing the service to the user, client application **1632** can receive request data from host application **1612** and provide user data in response to the request data. OTT connection **1650** can transfer both the request data and the user data. Client application **1632** can interact with the user to generate the user data that it provides. Software **1631** can also include program instructions (also referred to as a computer program product) that, when executed by processing circuitry **1638**, can configure UE **1630** to perform operations corresponding to various exemplary methods (e.g., procedures) described herein.

(158) As an example, host computer **1610**, base station **1620** and UE **1630** illustrated in FIG. **16** can be similar or identical to host computers or base stations described in relation to other to figures herein. For example, the inner workings of these entities can be as shown in FIG. **16** and independently, the surrounding network topology can be that shown in other figures herein.

(159) In FIG. **16**, OTT connection **1650** has been drawn abstractly to illustrate the communication between host computer **1610** and UE **1630** via base station **1620**, without explicit reference to any intermediary devices and the precise routing of messages via these devices. Network infrastructure can determine the routing, which it can be configured to hide from UE **1630** or from the service provider operating host computer **1610**, or both. While OTT connection **1650** is active, the network infrastructure can further take decisions by which it dynamically changes the routing (e.g., on the basis of load balancing consideration or reconfiguration of the network).

(160) Wireless connection **1670** between UE **1630** and base station **1620** is in accordance with the

teachings of the embodiments described throughout this disclosure. One or more of the various embodiments improve the performance of OTT services provided to UE **1630** using OTT connection **1650**, in which wireless connection **1670** forms the last segment. More precisely, the exemplary embodiments disclosed herein can improve flexibility for the network to monitor end-to-end quality-of-service (QoS) of data flows, including their corresponding radio bearers, associated with data sessions between a user equipment (UE) and another entity, such as an OTT data application or service external to the 5G network. These and other advantages can facilitate more timely design, implementation, and deployment of 5G/NR solutions. Furthermore, such embodiments can facilitate flexible and timely control of data session QoS, which can lead to improvements in capacity, throughput, latency, etc. that are envisioned by 5G/NR and important for the growth of OTT services.

(161) A measurement procedure can be provided for the purpose of monitoring data rate, latency and other network operational aspects on which the one or more embodiments improve. There can further be an optional network functionality for reconfiguring OTT connection **1650** between host computer **1610** and UE **1630**, in response to variations in the measurement results. The measurement procedure and/or the network functionality for reconfiguring OTT connection **1650** can be implemented in software **1611** and hardware **1615** of host computer **1610** or in software **1631** and hardware **1635** of UE **1630**, or both. In embodiments, sensors (not shown) can be deployed in or in association with communication devices through which OTT connection **1650** passes; the sensors can participate in the measurement procedure by supplying values of the monitored quantities exemplified above, or supplying values of other physical quantities from which software **1611**, **1631** can compute or estimate the monitored quantities. The reconfiguring of OTT connection **1650** can include message format, retransmission settings, preferred routing etc.; the reconfiguring need not affect base station **1620**, and it can be unknown or imperceptible to base station **1620**. Such procedures and functionalities can be known and practiced in the art. In certain embodiments, measurements can involve proprietary UE signaling facilitating host computer **1610**'s measurements of throughput, propagation times, latency and the like. The measurements can be implemented in that software **1611** and **1631** causes messages to be transmitted, in particular empty or 'dummy' messages, using OTT connection **1650** while it monitors propagation times, errors, etc.

(162) FIG. **17** is a flowchart illustrating an exemplary method and/or procedure implemented in a communication system, in accordance with one embodiment. The communication system includes a host computer, a base station and a UE which, in some exemplary embodiments, can be those described with reference to other figures herein. For simplicity of the present disclosure, only drawing references to FIG. **17** will be included in this section. In step **1710**, the host computer provides user data. In substep **1711** (which can be optional) of step **1710**, the host computer provides the user data by executing a host application. In step **1720**, the host computer initiates a transmission carrying the user data to the UE. In step **1730** (which can be optional), the base station transmits to the UE the user data which was carried in the transmission that the host computer initiated, in accordance with the teachings of the embodiments described throughout this disclosure. In step **1740** (which can also be optional), the UE executes a client application associated with the host application executed by the host computer.

(163) FIG. **18** is a flowchart illustrating an exemplary method and/or procedure implemented in a communication system, in accordance with one embodiment. The communication system includes a host computer, a base station and a UE which can be those described with reference to other figures herein. For simplicity of the present disclosure, only drawing references to FIG. **18** will be included in this section. In step **1810** of the method, the host computer provides user data. In an optional substep (not shown) the host computer provides the user data by executing a host application. In step **1820**, the host computer initiates a transmission carrying the user data to the UE. The transmission can pass via the base station, in accordance with the teachings of the

embodiments described throughout this disclosure. In step **1830** (which can be optional), the UE receives the user data carried in the transmission.

(164) FIG. **19** is a flowchart illustrating an exemplary method and/or procedure implemented in a communication system, in accordance with one embodiment. The communication system includes a host computer, a base station and a UE which can be those described with reference to other figures herein. For simplicity of the present disclosure, only drawing references to FIG. **19** will be included in this section. In step **1910** (which can be optional), the UE receives input data provided by the host computer. Additionally or alternatively, in step **1920**, the UE provides user data. In substep **1921** (which can be optional) of step **1920**, the UE provides the user data by executing a client application. In substep **1911** (which can be optional) of step **1910**, the UE executes a client application which provides the user data in reaction to the received input data provided by the host computer. In providing the user data, the executed client application can further consider user input received from the user. Regardless of the specific manner in which the user data was provided, the UE initiates, in substep **1930** (which can be optional), transmission of the user data to the host computer. In step **1940** of the method, the host computer receives the user data transmitted from the UE, in accordance with the teachings of the embodiments described throughout this disclosure.

(165) FIG. **20** is a flowchart illustrating an exemplary method and/or procedure implemented in a communication system, in accordance with one embodiment. The communication system includes a host computer, a base station and a UE which can be those described with reference to other figures herein. For simplicity of the present disclosure, only drawing references to FIG. **20** will be included in this section. In step **2010** (which can be optional), in accordance with the teachings of the embodiments described throughout this disclosure, the base station receives user data from the UE. In step **2020** (which can be optional), the base station initiates transmission of the received user data to the host computer. In step **2030** (which can be optional), the host computer receives the user data carried in the transmission initiated by the base station.

(166) The foregoing merely illustrates the principles of the disclosure. Various modifications and alterations to the described embodiments will be apparent to those skilled in the art in view of the teachings herein. It will thus be appreciated that those skilled in the art will be able to devise numerous systems, arrangements, and procedures that, although not explicitly shown or described herein, embody the principles of the disclosure and can be thus within the spirit and scope of the disclosure. Various exemplary embodiments can be used together with one another, as well as interchangeably therewith, as should be understood by those having ordinary skill in the art.

(167) The term unit, as used herein, can have conventional meaning in the field of electronics, electrical devices and/or electronic devices and can include, for example, electrical and/or electronic circuitry, devices, modules, processors, memories, logic solid state and/or discrete devices, computer programs or instructions for carrying out respective tasks, procedures, computations, outputs, and/or displaying functions, and so on, as such as those that are described herein.

(168) Any appropriate steps, methods, features, functions, or benefits disclosed herein may be performed through one or more functional units or modules of one or more virtual apparatuses. Each virtual apparatus may comprise a number of these functional units. These functional units may be implemented via processing circuitry, which may include one or more microprocessor or microcontrollers, as well as other digital hardware, which may include Digital Signal Processor (DSPs), special-purpose digital logic, and the like. The processing circuitry may be configured to execute program code stored in memory, which may include one or several types of memory such as Read Only Memory (ROM), Random Access Memory (RAM), cache memory, flash memory devices, optical storage devices, etc. Program code stored in memory includes program instructions for executing one or more telecommunications and/or data communications protocols as well as instructions for carrying out one or more of the techniques described herein. In some implementations, the processing circuitry may be used to cause the respective functional unit to

perform corresponding functions according one or more embodiments of the present disclosure.

(169) As described herein, device and/or apparatus can be represented by a semiconductor chip, a chipset, or a (hardware) module comprising such chip or chipset; this, however, does not exclude the possibility that a functionality of a device or apparatus, instead of being hardware implemented, be implemented as a software module such as a computer program or a computer program product comprising executable software code portions for execution or being run on a processor.

Furthermore, functionality of a device or apparatus can be implemented by any combination of hardware and software. A device or apparatus can also be regarded as an assembly of multiple devices and/or apparatuses, whether functionally in cooperation with or independently of each other. Moreover, devices and apparatuses can be implemented in a distributed fashion throughout a system, so long as the functionality of the device or apparatus is preserved. Such and similar principles are considered as known to a skilled person.

(170) Furthermore, functions described herein as being performed by a wireless device or a network node may be distributed over a plurality of wireless devices and/or network nodes. In other words, it is contemplated that the functions of the network node and wireless device described herein are not limited to performance by a single physical device and, in fact, can be distributed among several physical devices.

(171) In addition, certain terms used in the present disclosure, including the specification, drawings and exemplary embodiments thereof, can be used synonymously in certain instances, including, but not limited to, e.g., data and information. It should be understood that, while these words and/or other words that can be synonymous to one another, can be used synonymously herein, that there can be instances when such words can be intended to not be used synonymously. Further, to the extent that the prior art knowledge has not been explicitly incorporated by reference herein above, it is explicitly incorporated herein in its entirety. All publications referenced are incorporated herein by reference in their entireties.

(172) Unless otherwise defined, all terms (including technical and scientific terms) used herein have the same meaning as commonly understood by one of ordinary skill in the art to which this disclosure belongs. It will be further understood that terms used herein should be interpreted as having a meaning that is consistent with their meaning in the context of this specification and the relevant art and will not be interpreted in an idealized or overly formal sense unless expressly so defined herein.

(173) In addition, certain terms used in the present disclosure, including the specification and drawings, can be used synonymously in certain instances (e.g., “data” and “information”). It should be understood, that although these terms (and/or other terms that can be synonymous to one another) can be used synonymously herein, there can be instances when such words can be intended to not be used synonymously. Further, to the extent that the prior art knowledge has not been explicitly incorporated by reference herein above, it is explicitly incorporated herein in its entirety. All publications referenced are incorporated herein by reference in their entireties.

(174) Example embodiments of the techniques and apparatus described herein include, but are not limited to, the following enumerated embodiments: A1. A method for a user equipment (UE) to synchronize timing with a wireless network, the method comprising: receiving, from a first node serving a first cell in the wireless network, a first message that includes an absolute reference time and a corresponding event within downlink (DL) transmissions by the first node; determining whether the UE received, from the first node, a timing adjustment value from which the UE can estimate a propagation delay of the DL transmissions from the first node to the UE; and based on determining that the UE did not receive the timing adjustment value, performing one of the following operations: transmitting, to the network node, an indication that the UE did not receive the timing adjustment value; or declaring radio link failure (RLF) with respect to the first cell and establishing a connection with a second cell that is capable of providing the information included in the first message. A2. The method of embodiment A1, wherein the timing adjustment value is a

timing advance (TA) command that indicates an amount of timing adjustment that the UE should apply to UL transmissions. A3. The method of embodiment A2, wherein determining whether the UE received the timing adjustment value comprises determining whether the UE received a medium access control (MAC) control element (CE) comprising the timing adjustment value during one or more of the following: a most recent first duration; and a second duration after receiving the first message. A4. The method of any of embodiments A2-A3, further comprising: receiving, from the first node, a timer value associated with a timing advance group (TAG) that includes the first cell; and initiating a timer based on the timer value upon receiving a previous timing adjustment value from the first node, wherein determining that the UE did not receive the timing adjustment value is based on expiration of the timer. A5. The method of embodiment A4, wherein transmitting the indication that the UE did not receive the timing adjustment value is responsive to expiration of the timer. A6. The method of any of embodiments A4-A5, wherein declaring RLF is based on the timer being initiated and expiring a plurality of consecutive times without receiving the timing adjustment value. A7. The method of any of embodiments A2-A6, further comprising, based on determining that the UE did receive the timing adjustment value, selectively using the timing adjustment value for estimating the propagation delay of the DL transmissions. A8. The method of embodiment A7, wherein selectively using the timing adjustment value comprises using the timing adjustment value only when the value of a flag, received with the TA command in the MAC CE, is different from a most recent value of the flag received with a most recent TA command. A9. The method of any of embodiments A1-A8, wherein: the method further comprises sending, to the first node, an indication of a UE preference for receiving reference time information from the wireless network; and the first message is received in response to the indication. A10. The method of any of embodiments A1-A9, wherein establishing a connection with a second cell that is capable of providing the information included in the first message comprises: receiving information broadcast by the second cell; and determining that the broadcast information includes a field reserved for delivery of reference time information. A11. The method of any of embodiments A1-A9, wherein establishing a connection with a second cell that is capable of providing the information included in the first message comprises: sending, to a second network node serving the second cell, a reestablishment request message comprising a request for reference time information; and receiving, from the second network node, a reestablishment message comprising the requested reference time information. A12. The method of any of embodiments A1-A11, wherein the first node, the first cell, and the second cell are part of a standalone non-public network (SNPN). A13. The method of any of embodiments A1-A11, wherein: the first node is part of a public network integrated non-public network (PNI-NPN); the UE is part of a closed access group (CAG) associated with the PNI-NPN; and the first cell and the second cell are associated with the CAG. A14. The method of any of embodiments A1-A13, further comprising receiving, from the network node, a retransmission of the timing adjustment value after transmitting the indication. B1. A method for a network node, serving a cell in a wireless network, to synchronize timing with a user equipment (UE), the method comprising: transmitting, to the UE, a first message that includes an absolute reference time and a corresponding event within downlink (DL) transmissions by the network node; periodically transmitting, to the UE, respective timing adjustment values from which the UE can estimate a propagation delay of the DL transmissions from the first node to the UE; and based on determining that the UE did not receive one of the periodic transmissions of the timing adjustment value, retransmitting the non-received timing adjustment value to the UE. B2. The method of embodiment A1, wherein each timing adjustment value is: a timing advance (TA) command that indicates an amount of timing adjustment that the UE should apply to UL transmissions; and transmitted in a medium access control (MAC) control element (CE). B3. The method of embodiment B2, further comprising sending, to the UE, a timer value associated with a timing advance group (TAG) that includes the cell, wherein the timer value is related to a period between successive transmissions of respective timing adjustment values. B4.

The method of embodiment B3, wherein determining that the UE did not receive one of the periodic transmissions comprises receiving, from the UE, an indication that the UE did not receive the timing adjustment value during a duration corresponding to the timer value. B5. The method of any of embodiments B2-B4, wherein each timing adjustment value is transmitted in a MAC CE together with a flag. B6. The method of embodiment B5, wherein: a first value of the flag indicates that the timing adjustment value is different from a most recently transmitted timing adjustment value; and a second value of the flag indicates that the timing adjustment value is the same as the most recently transmitted timing adjustment value. B7. The method of any of embodiments B1-B6, wherein: the method further comprises receiving, from the UE, an indication of the UE's preference for receiving reference time information from the wireless network; and the first message is sent in response to the indication. B8. The method of any of embodiments B1-B7, wherein the network node and the first cell are part of a standalone non-public network (SNPN). B9. The method of any of embodiments B1-B7, wherein: the network node is part of a public network integrated non-public network (PNI-NPN); the UE is part of a closed access group (CAG) associated with the PNI-NPN; and the cell is associated with the CAG. B10. The method of embodiment B9, further comprising transmitting the first message in one or more further cells associated with the CAG. C1. A user equipment (UE) configured to synchronize timing with a wireless network, the UE comprising: radio interface circuitry configured to communicate with a network node in the wireless network; and processing circuitry operably coupled to the radio interface circuitry, whereby the processing circuitry and the radio interface circuitry are configured to perform operations corresponding to any of the methods of embodiments A1-A13. C2. A user equipment (UE) configured to synchronize timing with a wireless network, the UE being further arranged to perform operations corresponding to any of the methods of embodiments A1-A14. C3. A non-transitory, computer-readable medium storing program instructions that, when executed by processing circuitry of a user equipment (UE) configured to synchronize timing with a wireless network, configure the UE to perform operations corresponding to any of the methods of embodiments A1-A14. C4. A computer program product comprising program instructions that, when executed by processing circuitry of a user equipment (UE) configured to synchronize timing with a wireless network, configure the UE to perform operations corresponding to any of the methods of embodiments A1-A14. D1. A network node configured to serve a cell in a wireless network and to synchronize timing with a user equipment (UE) in the cell, the network node comprising: communication interface circuitry configured to communicate with the UE; and processing circuitry operably coupled with the communication interface circuitry, whereby the processing circuitry and the communication interface circuitry are configured to perform operations corresponding to any of the methods of embodiments B1-B10. D2. A network node configured to serve a cell in a wireless network and to synchronize timing with a user equipment (UE) in the cell, the network node being further arranged to perform operations corresponding to any of the methods of embodiments B1-B10. D3. A non-transitory, computer-readable medium storing program instructions that, when executed by processing circuitry of a network configured to serve a cell in a wireless network and to synchronize timing with a user equipment (UE) in the cell, configure the network node to perform operations corresponding to any of the methods of embodiments B1-B10. D4. A computer program product comprising program instructions that, when executed by processing circuitry of a network node configured to serve a cell in a wireless network and to synchronize timing with a user equipment (UE) in the cell, configure the network node to perform operations corresponding to any of the methods of embodiments B1-B10.

Claims

1. A method for a user equipment (UE) to synchronize timing with a wireless network, the method comprising: receiving, from a first node serving a first cell in the wireless network, a first message

that includes an absolute reference time and a corresponding reference event within downlink (DL) transmissions by the first node; determining whether the UE received, from the first node, a timing advance (TA) command that indicates an amount of timing adjustment that the UE should apply to uplink (UL) transmissions; and based on determining that the UE did not receive the TA command, performing one of the following operations: transmitting to the first node an indication that the UE did not receive the TA command; or declaring radio link failure (RLF) with respect to the first cell and establishing a connection with a second cell that is capable of providing the information included in the first message.

2. The method of claim 1, wherein determining whether the UE received the TA command comprises determining whether the UE received a medium access control (MAC) control element (CE) comprising a TA value during one or more of the following: a most recent first duration; and a second duration after receiving the first message.

3. The method of claim 2, further comprising, based on determining that the UE did receive the TA command, selectively using the TA value for estimating the propagation delay of the DL transmissions.

4. The method of claim 3, wherein selectively using the TA value comprises using the TA value only when the value of a flag, received with the TA command in the MAC CE, is different from a most recent value of the flag received with a most recent TA command.

5. The method of claim 1, further comprising: receiving from the first node a timer value associated with a timing advance group (TAG) that includes the first cell; and initiating a timer based on the timer value upon receiving a previous TA command from the first node, wherein determining that the UE did not receive the TA command is based on expiration of the timer.

6. The method of claim 5, wherein transmitting the indication that the UE did not receive the TA command is responsive to expiration of the timer.

7. The method of claim 5, wherein declaring RLF is based on the timer being initiated and expiring a plurality of consecutive times without receiving the TA command.

8. The method of claim 1, wherein: the method further comprises sending to the first node an indication of a UE preference for receiving reference time information from the wireless network; and the first message is received in response to the indication.

9. The method of claim 1, wherein establishing a connection with a second cell that is capable of providing the information included in the first message comprises: receiving information broadcast by the second cell; and determining that the received information includes a field reserved for delivery of reference time information.

10. The method of claim 1, wherein establishing a connection with a second cell that is capable of providing the information included in the first message comprises: sending, to a second node serving the second cell, a reestablishment request message comprising a request for reference time information; and receiving from the second node a unicast reestablishment message comprising the requested reference time information.

11. The method of claim 1, wherein the first node, the first cell, and the second cell are part of a standalone non-public network (SNPN).

12. The method of claim 1, wherein: the first node is part of a public network integrated non-public network (PNI-NPN); the UE is part of a closed access group (CAG) associated with the PNI-NPN; and the first cell and the second cell are associated with the CAG.

13. The method of claim 1, further comprising receiving, from the first node, a retransmission of the TA command after transmitting the indication.

14. A method for a first node serving a first cell of a wireless network to synchronize timing with a user equipment (UE) the method comprising: transmitting to the UE a first message that includes an absolute reference time and a corresponding reference event within downlink (DL) transmissions by the first node; periodically transmitting to the UE respective timing advance (TA) commands that indicate an amount of timing adjustment that the UE should apply to each UL

transmission; and based on determining that the UE did not receive one of the periodic transmissions of the TA commands, retransmitting the non-received TA command to the UE.

15. The method of claim 14, wherein the respective TA commands are transmitted in respective medium access control (MAC) control elements (CEs).

16. The method of claim 15, further comprising sending to the UE a timer value associated with a timing advance group (TAG) that includes the first cell, wherein the timer value is related to a period between successive transmissions of respective TA commands.

17. The method of claim 16, wherein determining that the UE did not receive one of the periodic transmissions comprises receiving from the UE an indication that the UE did not receive the TA command during a duration corresponding to the timer value.

18. The method of claim 15, wherein: each TA command is transmitted in a MAC CE together with a flag; a first value of the flag indicates that a timing adjustment value of the TA command is different from a timing adjustment value of the most recently transmitted TA command; and a second value of the flag indicates that the timing adjustment value of the TA command is the same as the timing adjustment value of the most recently transmitted TA command.

19. The method of claim 14, wherein: the method further comprises receiving from the UE an indication of the UE's preference for receiving reference time information from the wireless network; and the first message is sent in response to the indication.

20. The method of claim 14, wherein the first node and the first cell are part of a standalone non-public network (SNPN).

21. The method of claim 14, wherein: the first node is part of a public network integrated non-public network (PNI-NPN); the UE is part of a closed access group (CAG) associated with the PNI-NPN; and the first cell is associated with the CAG.

22. The method of claim 21, further comprising transmitting the first message in one or more further cells associated with the CAG.

23. A user equipment (UE) configured to synchronize timing with a wireless network, the UE comprising: communication interface circuitry configured to communicate with a first node in the wireless network; and processing circuitry operably coupled to the communication interface circuitry, whereby the processing circuitry and the communication interface circuitry are configured to: receive, from a first node serving a first cell in the wireless network, a first message that includes an absolute reference time and a corresponding reference event within downlink (DL) transmissions by the first node; determine whether the UE received, from the first node, a timing advance (TA) command that indicates an amount of timing adjustment that the UE should apply to UL transmissions; and based on determining that the UE did not receive the TA command, perform one of the following operations: transmit to the first node an indication that the UE did not receive the TA command; or declare radio link failure (RLF) with respect to the first cell and establish a connection with a second cell that is capable of providing the information included in the first message.

24. A first node configured to serve a first cell of a wireless network and to synchronize timing with a user equipment (UE), the first node comprising: communication interface circuitry configured to communicate with the UE; and processing circuitry operably coupled with the communication interface circuitry, whereby the processing circuitry and the communication interface circuitry are configured to perform operations corresponding to the method of claim 14.
